Hinode 19 Poster Program

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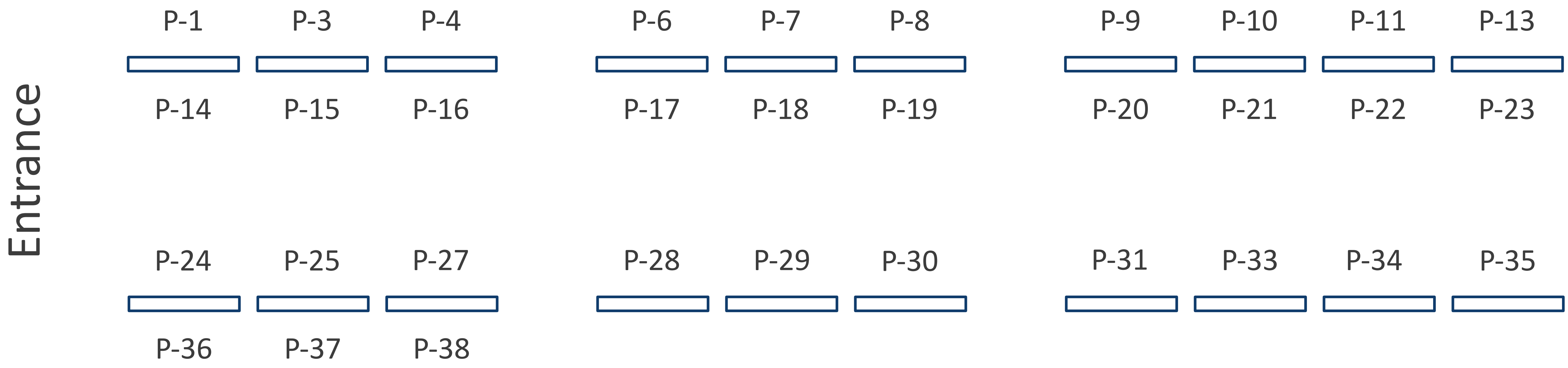
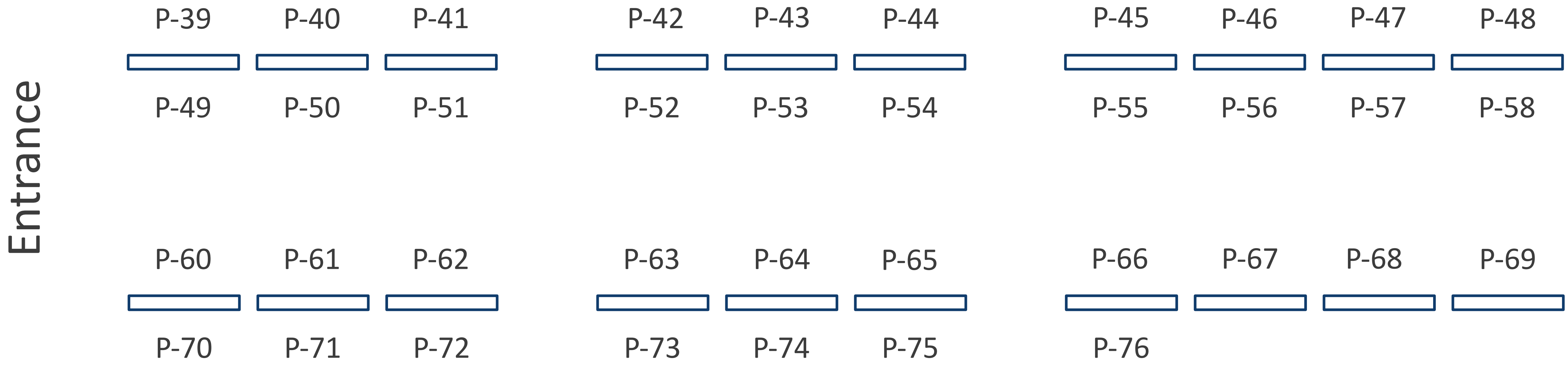
First Poster Session (Day 1 – Day 2)



First Poster Session (Day 3 – Day 5)



Overview of Solar EUV, UV & Soft X-ray Irradiance Variability and Their Impacts on Earth's Climate & Space Weather

*Kariyappa R.[1], Adithya H N[7], Masuda S.[2], Kusano K.[2], Imada S.[3], Zender J.[4], Dame L.[5], Deluca E. E.[6], Weber Mark[6]

1. Indian Institute of Astrophysics/ISEE, NU, 2. ISEE, Nagoya University, 3. Tokyo University, 4. ESA/ESTEC, 5. LATMOS/CNRS, 6. CFA, 7. MAHE

It is important to study the variabilities of solar EUV, UV and X-ray irradiance in heliophysics, in Earth’ s climate, and space weather applications. Since the radiative output of the Sun is one of the main driving forces of the terrestrial atmosphere and climate system, the study of solar energy has become of great interest and importance. Although the solar energy flux integrated over the entire spectrum is considered to be one of the major natural forces of the Earth’ s climate system, studying the extreme ultraviolet (EUV), ultraviolet (UV) and X-ray irradiance variability is particularly important in solar and terrestrial physics. The solar EUV, UV and X-ray fluxes play in particular a major role in the heating of the Earth’ s atmosphere and Solar-Terrestrial relationships. Thus it is an important issue to understand their variability and its applications in Earth’ s climate and space weather.

In this paper we discuss the variability of the total intensity, temperature and magnetic field of different coronal features such as active regions (ARs), coronal holes (CHs), X-ray bright points (XBPs) and background regions (BGs). The contribution of all these features to total energy flux over the full disk will be determined, including the total temperature and magnetic field. The magnetic field is the main source of all the surface features of the Sun. The role of magnetic field on the variability of intensity and temperature of the coronal features and the impacts of EUV/UV/X-ray irradiance variability on Earth's Climate & Space Weather is discussed in great detail in this paper.

ASPIICS coronagraph onboard the Proba-3 mission –observations and science

*Stanislav Gunar[1], Andrei Zhukov[2], Silvano Fineschi[3], Philippe Lamy[4], Hardi Peter[5], Marilena Mierla[6], Pawel Rudawy[7], Kanaris Tsinganos[8]

1. Astronomical Institute of the Czech Academy of Sciences, 251 65 Ondrejov, 2. Solar–Terrestrial Centre of Excellence —SIDC, Royal Observatory of Belgium, 1180 Brussels, 3. National Institute for Astrophysics, Astrophysical Observatory of Torino, Pino Torinese, Torino, 4. Laboratoire Atmospheres et Observations Spatiales, 11 Boulevard d' Alembert, 78280 Guyancourt, 5. Max Planck Institute for Solar System Research, Justus-von-Liebig-Weg 3, 37077 Gottingen, 6. Institute of Geodynamics of the Romanian Academy, 020032 Bucharest-37, 7. Astronomical Institute, University of Wroclaw, Kopernika 11, 51-622 Wroclaw, 8. University of Athens, Panepistimiopolis, 157 84 Zografos Athens

The giant distributed coronagraph ASPIICS onboard the formation-flying mission Proba-3 of ESA investigates the hitherto practically unexplored inner depths of the solar corona. This region lies above the reach of disk imagers such as SDO/AIA and below the inner limit of other space coronagraphs. Although difficult to observe, the inner corona is a place of great interest. This is where the fast solar wind gets accelerated to supersonic velocities and where CMEs also reach their maximum accelerations. It is also the place where the transition between the regions of the closed and open magnetic field often happens and the slow solar wind originates.

We will present the current status and the most recent scientific results obtained by the Proba-3 mission. We will also look at the unique concept of the formation-flying technology and how to coordinate observations with missions like Proba-3 with mode targeted, high-resolution but small FOV instruments.

Statistical Study of Flare Productivity in Newly Emerging Solar Active Regions

*Ayu Dyah Pangestu[1], Satoshi Masuda[1], Johan Muhamad[2]

1. Institute for Space-Earth Environmental Research, Nagoya University, 2. Research Center for Space, National Research and Innovation Agency (BRIN)

This study examines the flare productivity of newly emerging solar active regions (new ARs), distinguishing between isolated and those emerging in proximity to pre-existing active regions (PEARs). Both flaring and non-flaring cases are analyzed to assess the influence of the surrounding magnetic environment on flare occurrence in the new ARs. The dataset includes 1,582 new ARs identified from the NOAA catalog during the peak activity of Solar Cycles 24 (2010–2017) and 25 (2021–October 2025). Only new ARs emerging within heliographic longitudes E60 to W30 in the Stonyhurst coordinate system are considered. New ARs are classified as "paired" if they emerge within 12° of a PEAR, and "single" otherwise. Flare productivity is calculated using a flare index derived from solar soft X-ray flares of class C1.0 and above, observed within the first four days after emergence. Results indicate that paired new ARs are statistically more flare-productive than single ones. Peak flare productivity in paired ARs occurs at separations of approximately 8–10° from PEARs. Furthermore, new ARs emerging at higher latitudes relative to nearby PEARs tend to show increased flare activity. Additionally, the total X-ray intensity of new ARs is evaluated using segmentation maps from Hinode/XRT irradiance time series data. Intensity measurements also suggest slightly higher values for paired ARs. These findings highlight the role of the magnetic environment in modulating flare productivity, though differences remain modest. Future work will incorporate detailed coronal analyses to further study the underlying physical mechanisms.

Fast and periodic propagating disturbances along coronal loops detected with HRIEUV on board Solar Orbiter: a possible signature of MHD waves

*Antoine Dolliou[1], Sudip Mandal[1], Krzysztof Barczynski[2,3], Tom Van Doorsselaere[9], David Berghmans[4], Clara Froment[5], Frederic Auchere[6], Patrick Antolin[7], Henrik Eklund[8], Yingjie Zhu[3], Emil Kraaikamp[4], Francis Verbeeck[4]

1. MPS, 2. PMOD/WRC, 3. ETH-Zurich, 4. ROB, 5. LPC2E, 6. IAS, 7. Northumbria University, 8. ESA, 9. KU Leuven

We used observations of Solar Orbiter/Extreme Ultraviolet Imager (EUI) at high spatial (125 km) and temporal (5s) resolutions to investigate mass and energy transfer processes along active region (AR) coronal loops. To do so, we computed the time distance maps along 13 coronal loops. For six of them, we report the detection of Propagating disturbances (PDs) with high plane-of-sky (PoS) velocities along the loop (500 to 2000 km/s). They are only visible in the upper part of the loop, showing little to no intensity damping over their propagation path. In one case, we measured a period of two minutes with a Fourier analysis. These PDs show properties consistent with fast flows induced by magnetic reconnection, a heating front, or MHD waves. In particular, we performed analytical derivations and a 1.5D ideal MHD simulation to test whether fast PDs are consistent with a density perturbation caused by the ponderomotive force in an Alfvén wave. The origin of fast PDs also opens new perspectives, as they are likely to be located near the footpoints of the loop. This is why, in addition to being potential signatures of energy and mass transfer mechanisms, we suggest these PDs to be the signature of local and periodic energy depositions in the lower parts of the loops. We also discuss how future Solar-C/EUVST observations will help to estimate their contribution to coronal heating and as drivers of the solar wind.

3D NLTE Radiative Transfer with Factorized Fourier Neural Operators

*Harsh Mathur[1,2], Tiago Pereira[1,2]

1. Rosseland Centre for Solar Physics, University of Oslo, 2. Institute of Theoretical Astrophysics, University of Oslo

Three-dimensional non-local thermodynamic equilibrium (3D NLTE) radiative transfer is essential for interpreting modern solar and stellar atmosphere simulations, but conventional codes such as Multi3D remain computationally expensive for large snapshot sequences and repeated spectral synthesis. We present a Fourier Neural Operator-based framework designed as a geometry-aware surrogate to calculate 3D NLTE population calculations from 3D stellar atmospheres.
Unlike earlier machine-learning surrogates that use local neighborhoods to predict a center voxel or center column, our framework learns an operator from the full 3D atmospheric state to the full 3D NLTE departure coefficient volume. The model is therefore not a column-wise correction scheme, but a volumetric predictor trained directly on whole-atmosphere input-output fields. During training, it sees the surrounding atmospheric structure as part of the target itself, rather than using neighboring columns only as auxiliary context.
The inputs include temperature, density, electron density, and velocity components, while the targets are departure coefficients for selected atomic levels. The architecture combines horizontal spectral operators with a coordinate-conditioned vertical branch, enabling it to capture lateral radiative coupling and depth-dependent NLTE structure. Crucially, the model explicitly incorporates physical horizontal grid spacings dx and dy, and accepts a native z_scale, including arbitrary non-uniform vertical grids. It is therefore not tied to fixed pixel spacing, a uniform depth index, or a remapped common vertical coordinate.
This makes the framework a true volumetric surrogate for Multi3D: it preserves global atmospheric context, supports native simulation geometry, and outputs complete 3D departure coefficient cubes for downstream spectral synthesis.

Quiet Sun electron densities from the C III 97.7/117.6 line ratio

*Luca Nunzio Teriaca[1], Vincenzo Andretta[2]

1. Max Planck Institute for Solar System Research, 2. INAF-Osservatorio Astronomico di Capodimonte

In a fully ionized plasma, the measurement of the electron density provides a direct measurement of the mass density, a fundamental thermodynamical quantity. Perhaps the most reliable method to measure electron densities involves the ratio of spectral lines, or groups of lines, from the same ion but with substantially different spontaneous decay coefficients. These diagnostic ratios typically involve lines arising from forbidden transitions. Normally, however, forbidden lines are also very weak, posing therefore serious problems even to the study of slowly evolving phenomena. One of the most remarkable exceptions is the ratio of the permitted C III 97.7 nm line to the forbidden C III multiplet at 117.6 nm, both lines being very bright features of the solar VUV spectrum and both to be simultaneously observed by the EUVST spectrometer aboard SOLAR-C. However, the 117.6 nm multiplet is blended with autoionization lines from S I. Here we revise the atomic data for the S I autoionization lines by comparing laboratory measurements with solar data. Finally, we evaluate the importance of such a blend in diagnosing electron density in different solar regions and in solar-like stars, discussing in particular the usability of the C III 97.7/117.6 ratio under quiet Sun conditions.

Properties of main spectral lines observed by EUVST

*Regina Aznar Cuadrado[1], Antoine Dolliou[1], Luca Teriaca[1]

1. Max Planck Institute for Solar System Research

The EUVST spectrograph aboard Solar-C will observe simultaneously lines in one Short Wavelength (SW) band from 17 to 21.3 nm and in three Long Wavelength (LW) bands, LW1: 71.9 –84.7 nm, LW2: 92.8 –104.3 nm and LW3: 111.5 –122.1 nm. In addition, lines in the ranges 46.4 –52.15 nm and 55.75 –61.05 nm will be observed in the second order of diffraction in LW2 and LW3, respectively. In this work we use combined SOHO/SUMER and Hinode/EIS data to provide the main observables of the most relevant lines in these ranges, namely: the integrated line radiance, the line width (corrected from instrumental effects) and the non-thermal width in different regions of the Sun. This line catalogue is designed to be independent of the instrumental PSF, making it particularly useful to plan future Solar-C/EUVST observations.

New Opportunities for Coordinated Observations in Solar Orbiter’s High-Latitude Era

*Anik De Groof[1], David Williams[1], Catherine Fischer[1], Daniel Mueller[2], Miho Janvier[2]

1. European Spage Agency (ESAC), 2. European Space Agency (ESTEC)

Launched in February 2020, Solar Orbiter has delivered an extensive data collection to the scientific community, including high-resolution images and magnetic maps, detailed spectra, and in-situ measurements of the pristine solar wind from as close as 0.29au from the Sun. Its unique orbit around the Sun allows viewing our home star from a variety of different distances, latitudes and longitudinal separations with Earth. This in turn creates unique coordination opportunities with observatories on ground or close to Earth (Hinode, IRIS, SDO, Proba-3, Aditya-L1) and other deep-space missions like STEREO, BepiColombo and Parker Solar Probe.

Solar Orbiter’s high-latitude phase will lead to even more unique opportunities!In February 2025, a gravity-assist maneuver at Venus tilted the spacecraft’s orbit to 17° relative to the solar equator. This inclination will continue to increase over the coming years through additional Venus gravity assists, enabling Solar Orbiter to sample solar wind well beyond the ecliptic plane and investigate the Sun’s polar regions.In this contribution, we present the planned trajectory for the High Latitude Phase, highlighting the unique scientific opportunities for coordinated activities that this presents us with as we break free of the ecliptic view of our heliosphere.

Hard X-rays from CMEs

*Säm Krucker[1], Muriel Zoë Stiefel[1], Satoshi Masuda[2]

1. FHNW, 2. Nagoya University

Recent observations by Solar Orbiter STIX in combination with earth-orbiting observatories such as ASO-S/HXI have highlighted the diagnostic power of hard X-rays in detecting the hottest plasmas in CMEs. While hard X-rays from CMEs are faint, typically a few percent of the total flare flux, they reveal that the hottest CME plasmas have temperatures similar to the flare loops, reaching 30 MK in some cases. Due to their large spatial extent compared to the flare loop volume, this hot source within the CME can contain as much thermal energy as the flare loops themselves.

Temporal evolution of coronal voids

*Jonathan Nölke[1], Sami Solanki[1], Johann Hirzberger[1], Peter Hardi[1,2], Lakshmi Pradeep Chitta[1]

1. Max Planck Institute for Solar System Research, 2. Institute of Solar Physics (KIS)

Coronal voids are low-emission regions in the quiet Sun corona, first identified in Solar Orbiter EUI observations at around 1 MK. They typically span a few supergranular cells and exhibit intensity reductions of roughly 30% relative to the surrounding quiet Sun. Unlike coronal holes, coronal voids are magnetically closed features that are characterised by a significantly lower magnetic flux density and a largely absent magnetic network structures. This leads to weaker coronal heating and thus to the reduced emission.

In this work we investigate the temporal evolution of coronal voids and how their different physical properties relate to each other. By adapting the intensity-threshold method developed for EUI data, we identified and tracked coronal voids in the 171 A channel of SDO/AIA. This allowed us to analyse an extended quiet-Sun time series during solar minimum before Solar Orbiter’ s launch.

We studied the lifetimes and sizes of coronal voids, together with their magnetic flux density and emission across different coronal temperatures, and analysed correlations between these physical properties. We find that most coronal voids are short-lived features with typical lifetimes of only a few hours or less, which are considerably shorter than supergranular timescales. Furthermore, larger voids tend to persist longer, indicating a strong correlation between size and lifetime.

Recent Activities of Hinode Science Center at Nagoya

*Satoshi Masuda[1], Hinode Science Center at Nagoya

1. ISEE, Nagoya University

The Hinode Science Center was established at ISEE, Nagoya University in 2012. It promotes solar research by using precise data from Hinode and developing databases and analysis environment in collaboration with NAOJ and ISAS/JAXA. It continues to maintain, manage, and publish a list of dissertations employing data from Hinode (more than 100 Doctoral and about 70 Master' s theses), a flare event catalogue (including more than 30,000 events since October 2006), a 3D magnetic-field database of active regions in the corona, a polar magnetic-field database, an EUV emission line full-disk mosaic database, and a soft X-ray/EUV irradiance variability database. In addition to them, two databases related to solar radio observations were newly released in 2025. It is noteworthy that data DOI was given to each database and is useful as reference how each database is used for research in the solar physics community.Additionally, to enhance scientific outputs from the SOLAR-C mission, we are promoting numerical modeling studies of non-equilibrium ionization plasmas in the solar outer atmosphere and so forth.

Chromospheric Ca II 8542 ÅSignatures of Alfvén Waves Energy Deposition in an X-Class Solar Flare

*Teodora Mihailescu[1], Mariarita Murabito[1], Ilaria Ermolli[1]

1. INAF Osservatorio Astronomico di Roma

In solar flares, magnetic energy is released at high altitudes in the corona, transported downwards and dissipated as heat in the chromosphere, where it drives upwards chromospheric evaporation flows. Various physical mechanisms have been proposed to be responsible for this flare energy transport (particle beams, Alfvén waves, thermal conduction), and identifying ways to distinguish between these processes in observations is a key goal in solar flare physics. In this work, we study the link between chromospheric and coronal processes in an X-class solar flare. We focus on signatures of chromospheric processes observed in the Ca II 8542 Åline with the Interferometric Bidimensional Spectropolarimeter (IBIS) at the Dunn Solar Telescope (DST) and properties of the associated chromospheric evaporation flows inferred from Doppler shifts of hot flare lines lines observed by the Hinode EUV Imaging Spectrometer (EIS). We analyse different areas in the flaring region and assess the feasibility of using these signatures as a diagnostic for small-scale spatial variations in flare energy transport mechanisms.

Magnetic Suppression of Entropy-Rain Heat Transport in the Solar Convection Zone

*Rikuta Izutani[1], Hideyuki Hotta[1]

1. Nagoya Univ.

The amplitudes of large-scale convective flows in global simulations of the solar convection zone are substantially larger than those inferred from helioseismic observations. One possible way to resolve this “convective conundrum” is the entropy-rain scenario, in which small, low-entropy, dense downflows generated near the photosphere transport heat non-locally and reduce the need for vigorous giant-cell convection. While previous hydrodynamic studies have suggested that such localized thermal structures can propagate as coherent vortex-ring-like downflows, the role of magnetic fields, which are essential in solar and stellar convection zones, remains unclear.We investigate the magnetic influence on entropy-rain-like heat transport using two-dimensional compressible hydrodynamic and magnetohydrodynamic simulations in a gravitationally stratified atmosphere. A localized dense, low-entropy perturbation is introduced near the upper layer and evolves into a descending coherent structure. We compare hydrodynamic and magnetized cases by tracking the entropy minimum and by measuring the associated enthalpy flux, together with temperature perturbations and vertical velocities.Our analysis shows that the inclusion of a magnetic field systematically reduces the enthalpy flux carried by the descending structure. The reduction appears to be associated not only with changes in the downward velocity, but also with modifications of the entropy/temperature perturbation. These results suggest that magnetic fields can suppress the efficiency of entropy-rain heat transport, with important implications for modeling energy transport in the interiors of the Sun and cool stars.

Solar Flare Prediction Using Pre-training with an Autoencoder Architecture

*Kosuke Asatsuma[1], Yusuke Iida[1], Tomoki Kubo[1]

1. Niigata Univ.

In recent years, solar flare prediction using deep learning has been actively studied. Deep learning models require large training datasets to achieve high generalization performance since they have large number of trainable parameters. However, the available number of observational data of solar flares, particularly for X-class and M-class, are limited since they are natural phenomena, and it can lead unstable training or overfitting. To address this issue, we compare the flare prediction results of the CNN-LSTM architecture with and without pre-training using a Convolutional AutoEncoder (CAE).

We used approximately 140,000 line-of-sight magnetic field images of active regions observed by SDO/HMI from 2010 to 2017. M-class or larger flares are our targets of prediction. The training procedure consisted of two stages with pre-training. In the first stage, a CAE was first trained to learn feature representations of magnetic field images. Then, in the second stage the pre-trained encoder was incorporated into the CNN component of the CNN-LSTM model, and the entire model was trained in the end-to-end manner with flare occurrence. The performance was evaluated using a 5-fold cross-validation partitioned by active region.

Experimental results showed that the pre-training improved the generalization performance to TSS = 0.762±0.037, compared with TSS = 0.727±0.030 without pre-training. In addition, the pre-trained model showed more stable learning in the training. These results suggest that CAE-based pre-training effectively contributes to improve generalization performance and training stability in the flare prediction model.

CME Kinematics with Optical Flow: Multi-Coronagraph Insights into Internal Velocity Dispersion

*Pritam Das[1], Vaibhav Pant[2]

1. Aryabhatta Research Institute of Observational Sciences, 2. Indian Institute Of Technology Delhi

Coronal mass ejections (CMEs) result from the restructuring of the coronal magnetic field and the motion of the plasma, yet their internal velocity distributions remain poorly understood. In this work, we introduce an optical flow-based method, DOFCAT (Dense Optical Flow CME Analysis Tool), applied to high-resolution data from ASPIICS/Proba-3 and METIS/Solar Orbiter to examine the internal velocity dispersion of different substructures within CMEs in the Middle Corona. We also introduce a Gaussian-tapered Fourier (GTF) filter that reduces brightness flickering in running difference images from the ASPIICS wideband channel. Moreover, we corroborated our processing output with EUV data, such as SUVI and FSI, for validating the accuracy of DOFCAT. We then apply the optical flow technique to five CME events observed across ASPIICS and METIS, extracting their internal velocity information. Our results show that the velocity profiles of both the CME front and core change with position angle and are non-uniform. For impulsive CMEs, the height offset between the front and core begins to increase following the impulsive acceleration phase, while CMEs without a clear impulsive phase show a more gradual evolution of the height offset. We find that this increase in height difference occurs rapidly between 2.3-3.0 solar radii. Furthermore, the front shows a higher velocity spread than the core, with flanks contributing to lower velocity magnitudes, and impulsiveness evident in the CME leading half. These findings confirm that CMEs contain complex internal flows and layered substructures within the traditional three-part morphology. Our results demonstrate the utility of the optical flow technique on high-resolution coronagraphs for probing CME dynamics and provide new insights into CME initiation and evolution. Additionally, this work lays the foundation for a physics-based, automated CME detection pipeline using OF velocity distributions, a powerful alternative to traditional image-based methods for space weather forecasting.

Re-examining Hinode/XRT's Te and EM analysis : improved corrections of stray light and contamination on filters

*Aki Takeda[1]

1. Montana State University

The X-ray telescope onboard the Hinode satellite (Hinode/XRT) is a grazing incidence telescope staying in orbit since September 2006. The XRT equips 9 broad-band X-ray analysis filters, which enable us to perform temperature(Te) and emission measure(EM) analysis through the filter-ratio method assuming the isothermal plasma along the line of sight. The filter-ratio Te and EM calculated from the full-Sun integrated signals are used to synthesize the isothermal coronal spectrum to calculate the energy emitted in a specified wavelength range in the physical unit of W/m^2, i.e., X-ray irradiance. The solar X-ray irradiance obtained for nearly 2 solar activity cycles as well as a series of full-Sun images used to derive it constitute a valuable infrastructure for the the solar research community.

Through nearly 20 years of operation in space, however, XRT instruments have been suffering deterioration, which should be properly corrected for long-term quantitative data products: the visible light contamination from ruptures of pre-filter and the accretive contaminant layers on CCD and X-ray filters have been empirically corrected. Although this way of correction provides reasonable curve of solar irradiance, the determination of contaminant layer thickness was just arbitrary. We recently revised the method to make the correction more accountable: The effect of pre-filter ruptures was reexamined and the contaminant layer thickness on X-ray filters was introduced on the basis of frequency of filter use.

Influences of Magnetic Fields on Plume-Driven Solar Convection

*Kazuma Saito[1], Hideyuki Hotta[1], Allan Sacha Brun[2]

1. Nagoya Univ., ISEE, 2. CEA Paris-Saclay

The thermal convection in solar-type stars is thought to be driven by a local entropy gradient. Such a theory predicts the existence of convection-zone-scale convection (giant cells). However, giant cells have not yet been observationally confirmed in the Sun. This discrepancy between theory and observation is known as the convection conundrum, suggesting the need to revise current solar and stellar modeling.

As a possible solution to this problem, a hypothesis that has attracted attention in recent years proposes that convection in the Sun is driven not by local entropy gradients, but by strong downflows (plumes) generated near the solar surface, which is known as plume-driven convection. In this scenario, the entropy gradient in the bulk of the convection zone approaches zero, suppressing the driving of giant cells.

Several previous studies have performed numerical simulations of plume-driven convection. However, all of these studies neglected magnetic fields, leading to models that do not fully reflect the conditions in the convection zones of real stars, where self-consistent dynamo generated magnetic fields are known to play a key dynamical role. In this study, we investigated the effects of such dynamo-driven magnetic fields on plume-driven convection by conducting high performance numerical simulations in a Cartesian box.

The results show that the presence of magnetic fields tends to make the computational domain more convectively stable and reduces the kinetic energy of large-scale convection. In particular, the kinetic energy spectrum at large scales is reduced by approximately 10% compared to the non-magnetic case. These results provide support for the plume-driven convection hypothesis.

Statistical Study of Microflares Detected by Using Machine Learning Technique from NoRH Microwave Observation Data

*Ayatoshi Taniguchi[1], Satoshi Masuda[1], Yusuke Iida[2]

1. Nagoya University, 2. Niigata University

Heating mechanism of the solar corona remains unclear. Thus, this problem is one of primary scientific objectives of the SOLAR-C mission. One of the proposed models is micro/nanoflare heating model. To estimate the contribution of micro/nanoflares to coronal heating , the shape (or slope) of the flare frequency distribution is crucially important. While previous studies discussed distributions using EUV and X-ray observations, statistical results covering a wide energy range have not been fully established due to limitations such as short observational periods and small fields of view.
To overcome these limitations, we promote a statistical study of microflares using Nobeyama Radioheliograph (NoRH). This instrument is more sensitive to detect microflares in non-thermal emissions compared to conventional hard X-ray observations. Moreover, NoRH offers distinct advantages for statistical analysis with one-second temporal resolution, full-sun images, and long-term observation data spanning over 25 years.
The official NoRH flare list covers only relatively large solar flares, not including the small-scale flares targeted in this study. This means that it is necessary to develop a dedicated detection algorithm for microflares. Therefore, we developed a new automated detection algorithm specialized for microflares by applying the YOLO deep-learning object detection model to NoRH 17 GHz full-sun images. Our analysis of 25 years of data from 1993 to 2017 revealed that the resulting peak flux frequency distribution follows a power-law distribution in a wide flux range. We also derived the long-term variation of the frequency distribution over the 25-year period.
In addition to its implications for coronal heating research, the events detected in this study are valuable for investigating particle acceleration processes in microflares. We will also present preliminary results from an event analysis incorporating 34 GHz data.

Forward modeling of the Lyman line series for SOLAR-C

*Semya Amouche Tønnessen[1], Tiago M. D. Pereira[1]

1. Rosseland Centre for Solar Physics, University of Oslo

SOLAR-C/EUVST will observe the hydrogen Lyman series in unprecedented detail, covering six lines from Ly-α to Ly-ζ. To interpret and guide these future observations we employ forward modeling to study the formation and diagnostic potential of these lines. Making use of 3D radiative MHD simulations run with the Bifrost code, we study the impact of different physical ingredients in the formation of the lines, and how to extract physical quantities from their spectra. In this process, we deal also with how to make their spectral synthesis computationally tractable while retaining the necessary physical realism to model the profile shapes. The lines are not only influenced by departures from LTE, but can also be affected by partial redistribution (PRD), 3D effects, and linear Stark broadening in regimes far from the widely used recipe of Sutton (1978). We map the formation properties of the lines on a dynamic quiet Sun atmosphere, and discuss how one can combine the different Lyman lines to diagnose velocities and heating in the chromosphere.

How Small-scale Jet-like Solar Events from Miniature Flux Rope Eruptions Might Produce the Solar Wind

*Alphonse Sterling[1], Navdeep Panesar[2,3], Ronald Moore[4,1]

1. NASA/MSFC, 2. SETI Institute, 3. Lockheed Martin Solar and Astrophysics Laboratory, 4. University of Alabama Huntsville

We consider how small-scale jet-like events might make the solar wind, as has been suggested in recent studies. We show that the events referred to as “coronal jets” and as “jetlets” both fall on a power-law distribution that also includes large-scale eruptions and spicule-sized features; all of the jet-like events could contribute to the solar wind. Based on imaging and magnetic field data, it is plausible that many or most of these events might form by the same mechanism, in this manner: Magnetic flux cancelation produces small-scale flux ropes, often containing a cool-material minifilament; this minifilament/flux rope then erupts and reconnects with adjacent open coronal field, along which “plasma jets” flow and contribute to the solar wind. The erupting flux ropes can contain twist that is transferred to the open field and plausibly become Alfvénic pulses that form magnetic switchbacks, providing an intrinsic connection between switchbacks and the production of the solar wind. A summary of this work appears in Sterling, Panesar, & Moore (2024, ApJ, 963, 4).

Survey of small amplitude waves in solar prominences

*Kiyoshi Ichimoto[1,2], Ayumi Asai[2], Takako Ichii[2], Satoru UeNo[2]

1. Ritsumeikan University, 2. Kyoto University

Presence of small amplitude oscillation or waves in solar prominences was investigated by using archived data from a full disk Hα imaging spectroscopic observation. A total of 178 prominences on the limb were studied and it was found that there are plenty of evidence of oscillations
or waves with preferential periods in 3 –5 min (4 min) and 10 –20 min (15 min). We found very clear signature of persistently propagating 4 min wave in 21 prominences and at least a hint of 4 min oscillation in 90 prominences in their Doppler signal. The 4 min waves are often found in prominences with a large horizontal size, i.e., long filaments seen from a side, and they propagate in nearly horizontal direction with a phase speed of 140 –400 km/s. On the other hand, very clear evidence of the 15 min wave was found in 14 prominences and at least a hint of oscillation in 62 prominences in the Doppler signal. The 15 min waves are preferentially propagate in vertical direction with a phase speed of 20 –50 km/s. The 4 min waves are interpreted as Alfvénic wave propagating along the coronal magnetic field supporting the prominences and they are excited in external sources. Prominences that harbor the short period waves or oscillations distribute over the entire solar latitude. This
supports the picture that these short period waves are excited by ubiquitous disturbance in solar atmosphere. Our study revealed that persistently propagating short period waves in prominences is not particularly rare but rather common phenomena, and suggests that observing the 4 min waves in prominences will serve as a useful tool for tracing the coronal Alfvénic waves and the magnetic field supporting the prominences. The result of this study implies that the magnetic fields are preferentially horizontal in long quiescent prominences.

Comparison between Sun-as-a-star and Spatially Resolved Velocities of Solar Prominence Eruptions

*Takato Otsu[1,2], Ayumi Asai[2]

1. ISAS/JAXA, 2. Kyoto University

Hα spectra of stellar flares often exhibit Doppler-shifted components, which are sometimes interpreted as signatures of prominence/filament eruptions. However, the physical meaning of velocity parameters derived from Gaussian fitting, which is commonly used for analyzing such Doppler shifted signatures, remains unclear due to the lack of spatial information in stellar data. In this study, we investigate the physical meaning of velocity parameters (central velocity and velocity width) in Gaussian fitting using spatially resolved Hα data of solar prominence eruptions. We performed cloud model fitting of Hα spectra to derive Doppler velocities at each spatial position and examined their distributions. From these velocity distributions, we derived the spatially averaged velocity and the width of the velocity distribution, which represent the overall motion and the untwisting/expanding motions of prominences, respectively. In parallel, we constructed Sun-as-a-star Hα spectra by spatially integrating the Hα spectra and applied Gaussian fitting to derive the corresponding central velocity and velocity width. We find that the Gaussian central velocity closely corresponds to the spatially averaged velocity of the prominence, indicating that the central velocity traces the overall motion of erupting structures. Furthermore, the Gaussian velocity width is well approximated by the quadratic sum of the width of the spatial velocity distribution and the spatially averaged Doppler width. This implies that not only the overall motion but also the untwisting/expanding motions of erupting prominences can be inferred from spatially unresolved stellar spectra. Our results are further supported by an analytical formulation based on simplified assumptions. This study provides a robust physical basis for interpreting stellar spectra.

Flare EUV and Soft X-ray Contributions to the magnetic Solar Flare Effect via Global Geomagnetic Analyses

*Ryosuke Okubo[1], Kyoko Watanabe[1], Satoshi Masuda[2], Akimasa Ieda[2], Hidekatsu Jin[3], Chihiro Tao[3]

1. National Defense Academy of Japan, 2. ISEE/Nagoya university, 3. NICT

The Solar Flare Effect (SFE) is a spike-like geomagnetic variation driven by suddenly enhanced ionospheric currents following solar flares. To understand the correspondence between solar emissions (particularly soft X-ray and extreme ultraviolet (EUV)) and the ionospheric response, we investigated the spatiotemporal evolution of the SFE using global magnetometer networks.

Using global data (e.g., INTERMAGNET, SuperMAG) during X-class flares (2011–2014) when SDO/EVE was operational, we examined horizontal (X-Y) geomagnetic hodograms and the spatial distribution of equivalent current vortices by deriving vector differences between solar quiet (Sq) and SFE currents. Global hodogram analyses revealed abrupt changes in SFE vectors during flare progression. These turning points in the hodogram slopes synchronized with the transition of dominance between EUV (e.g., He II and Lyman-β) and soft X-ray fluxes. Furthermore, the global mapping of the vector differences between Sq and SFE vortices demonstrated that the SFE current forms an independent equivalent current system, distinct from Sq current, varying with latitude and local time.

We attribute both the hodogram directional shifts and the spatial divergence of the current vortices to the altitude-dependent ionospheric conductivities. Soft X-rays primarily ionize the lower E and D regions, enhancing Hall conductivity (peaking at ~105 km), whereas EUV radiation ionizes the upper E region, enhancing Pedersen conductivity (~130 km) (Richmond and Thayer, 2000). Because the temporal profiles of EUV and soft X-rays differ, the balance between Hall and Pedersen conductivity dynamically changes during a flare. By integrating global hodograms and current vortex analyses, this study highlights that the SFE is a dynamic superposition of multiple altitude-dependent current systems driven by distinct solar spectral bands.

The EUV spectroscopic observations by the upcoming SOLAR-C mission will be crucial for precisely determining the spectral evolution of flares and further elucidating these complex ionospheric responses.

Particle Acceleration Diagnostics in Preparation for the Solar-C Mission: Stereoscopic Non-thermal Flare Observations and 3D Magnetohydrodynamic Modeling of the 2024 October 1 X7.1 Flare

*Keitarou Matsumoto[1,2], Satoshi Inoue[2], Meiqi Wang[2], Bin Chen[2], Muriel Zoë Stiefel[4], Säm Krucker[3,5], Satoshi Masuda[1], Haimin Wang[2], Jeongwoo Lee[2]

1. Institute for Space-Earth Environmental Research, 2. New Jersey Institute of Technology, 3. University of Applied Sciences and Arts Northwestern Switzerland, 4. ETH Zürich, 5. University of California

Solar active region 13842 produced an X7.1-class flare on 2024 October 1. Although this event was extensively observed by modern solar missions, the underlying three-dimensional (3D) magnetic field structure cannot be directly measured, limiting our ability to fully understand the associated magnetic dynamics, particle acceleration, and transport. In this study, we reconstructed the 3D dynamics of the magnetic fields using data-constrained Magnetohydrodynamic (MHD) simulations, with a nonlinear force-free field as the initial condition. We then compared it with stereoscopic non-thermal emission from Solar Orbiter/STIX, ASO-S/HXI, and EOVSA.

The simulation reproduces reconnected flare loops consistent with the observed conjugate HXR footpoints, and the stereoscopically constrained height of the non-thermal coronal source agrees with the modeled looptop height. These results demonstrate that data-constrained MHD modeling can provide a realistic 3D magnetic context for interpreting non-thermal observations. Additional diagnostics, including the secondary microwave source, the migration of HXR footpoint sources within the same quasi-separatrix layer system, and the magnetic field strength inferred from microwave spectral fitting, further support a close connection between the observed non-thermal emissions and the evolving 3D magnetic structure in the simulation.

Recent observations from instruments and missions such as STIX, HXI, EOVSA, and FOXSI have substantially improved observational constraints on particle acceleration in solar flares. In this context, we discuss how the Solar-C mission can contribute to current and next-generation non-thermal observations by providing complementary diagnostics of the dynamic thermal plasma surrounding acceleration sites and by enabling more physically realistic data-constrained and data-driven MHD simulations.

Global Structure and Long-term Evolution of the Solar Wind: Results from 50 Years of IPS Observations in Japan

*Ken'ichi Fujiki[1], Kazumasa Iwai[1]

1. Institute for Space-Earth Envirometal Research (ISEE), Nagoya University

Interplanetary scintillation (IPS) observations offer a powerful remote-sensing technique for probing the solar wind by measuring intensity fluctuations of cosmic radio sources caused by the solar wind plasma. In this study, we present results from continuous IPS observations conducted in Japan over more than 50 years, using multiple radio arrays operated by the Institute for Space-Earth Environmental Research, Nagoya University. These long-term observations enable reconstruction of the global solar wind structure and its temporal evolution.
The IPS data, spanning both 70 MHz and 327 MHz observation systems, have been used to derive solar wind speed maps through tomographic analysis. While early data suffered from limited radio source coverage and data gaps, particularly during the 1970s, subsequent advancements in instrumentation and the optimization of observation schedules have significantly enhanced data continuity and spatial resolution. Furthermore, the integration of Japanese IPS data with datasets from other institutions has further strengthened the reliability of long-term analyses.
Our results reveal that the large-scale solar wind structure evolves smoothly over solar cycles, with the boundary between fast and slow solar wind varying in accordance with the tilt angle of the heliospheric current sheet. Despite limitations in reconstructing the southern hemisphere structure in earlier observations, the overall consistency between datasets demonstrates the robustness of IPS as a diagnostic tool.
This study highlights the importance of long-term IPS observations in understanding the global and long-term behavior of the solar wind, and further emphasizes the need to develop improved reconstruction techniques for solar wind velocity structures from 1970s observations to establish a more reliable global solar wind database.

Flow Patterns around Starspots in Low-Mass Stars

*Sangjin Park[1], Hideyuki Hotta[1]

1. Nagoya Univ.

We investigate the penumbral structure of starspots on low-mass stars using radiative magnetohydrodynamic (RMHD) simulations for 0.3M, 0.6M, and 1.0M, where M is the solar mass. While the 1.0M model reproduces solar-like penumbrae with filamentary structures and outward Evershed flows, the lower-mass models exhibit fundamentally different behavior. In the 0.3M and 0.6M cases, the penumbra lacks solar-like filamentary structures and overturning convection typical of sunspots, and instead shows strong inward flows with enhanced downflows near the outer penumbral boundary. These flows form large-scale converging patterns around the spot and do not show a clear correlation between upflow and intensity, as seen in the solar case. Mass-flux analysis reveals that the low-mass cases are characterized by net converging flows toward the spot, in contrast to the diverging flow pattern in the 1.0M model. Our results indicate that the mechanisms responsible for solar penumbral formation and the Evershed flow do not operate in the same way in low-mass stars.

Updates on the development of an LTE EoS for MPI-AMRVAC

*Jack Michael Jenkins[1], Christopher Osborne[2], Ben Snow[3]

1. European Space Agency, ESAC, 2. The University of Glasgow, 3. The University of Exeter

Solar atmospheric layers between roughly 4 and 20 kK -chromosphere, transition region, prominence cores, and the cool ends of coronal-rain blobs - sit on the hydrogen and helium ionisation knees, where ionisation degree and the first adiabatic exponent (γ_1) drop well below the below the fully-ionised defaults (A_He = 0.1: n_e/n_H ≈1.2, γ_1 = 5/3) used in corona-focused, fully-ionised (M)HD codes. We have implemented a Saha LTE equation of state in MPI-AMRVAC, together with cooling-operator changes (variable specific heat) and well-balanced reconstruction needed to use it in stratified production runs. This talk reports on the development effort, our validation strategy, and early science results. "Tabulated EoS" shock tests reproduce both exact Riemann solutions and two-γ analytical limits; standing sound-wave tests reproduce the analytic Saha γ_1 across the full chromospheric temperature range; quiet hydrostatic atmosphere benchmarks demonstrate long-term equilibrium preservation; and controlled FI/LTE comparison runs in 1D loops, ND prominences, and multi-dimensional flare configurations let us pinpoint where partial ionisation changes the dynamics. We highlight the regimes where the LTE closure matters most (e.g., slow-mode propagation in cool plasmas, condensation-onset timing, and the latent-heat buffer during catastrophic cooling). We close with the status of the public implementation and what we see as the natural next steps for making it available to the broader community.

Inspecting supersonic flows in granules: Stereoscopic study using Hinode/SP and Solar Orbiter/PHI

Saida Díaz Castillo[1,2], Takayoshi Oba[3], *Catherine Fischer[4], Luis Bellot Rubio[5], Nikola Vitas[6]

1. French-Spanish Laboratory for Astrophysics in Canaries (FSLAC), THEMIS S.L.U., 2. Institut für Sonnenphysik (KIS), 3. Advanced Research Center for Space Science and Technology (ARC-SAT), Institute of Science and Engineering, Kanazawa University, 4. European Space Agency, ESAC, 5. Instituto de Astrofísica de Andalucía (IAA-CSIC), 6. Instituto de Astrofísica de Canarias (IAC)

Solar granulation is the visible signature of overshooting convection, where cells of hot gas emerge from the upper layers of the convection zone into the solar photosphere. The convective dynamics within the photosphere have been extensively studied through both observations and realistic simulations. One of the most remarkable findings from these simulations is the predicted existence of supersonic horizontal flows within granules, which are estimated to cover approximately 4% of the solar surface at any given time. Spectropolarimetric studies of quiet-Sun granules have confirmed these supersonic flows and show that their visibility increases toward the solar limb. Despite these findings, deriving the full velocity vector of such structures remains challenging. This difficulty arises because measurements are typically limited to the line-of-sight(LOS) velocity; furthermore, inferring the apparent motion of horizontal plasma is complicated by diverse foreshortening and projection effects. Stereoscopic observations offer a promising strategy to overcome these hurdles. In particular, the combined capabilities of the Solar Orbiter/Polarimetric Helioseismic Imager (SO/PHI) and the Hinode/Solar Optical Telescope-Spectropolarimeter (SOT/SP) are ideal for this purpose.In this contribution, we present preliminary results from a study of horizontal supersonic flows in granules using coordinated observations from both instruments. Using a precisely co-aligned dataset, we identify regions with complex spectral signatures, evidencing multiple components in the blue wing of the spectra, and analyse the LOS velocities inferred by both instruments while accounting for the unique geometrical configuration of the spacecrafts. By leveraging the high spatial and temporal resolution afforded by this combination, we characterise the proper-
ties of granulation associated with supersonic flows and compare these findings with realistic simulation results.

Solar Atmospheres Without Compromise: Next-Generation Bifrost on the Dispatch Framework

*Mikolaj Marcin Szydlarski[1], Mats Carlsson[1], Viggo Haraldson Hansteen[1,2,3], Boris Vilhelm Guidiksen[1], Andrius Popovas[1], Åke Nordlund[4]

1. RoCS, ITA, University of Oslo, 2. Lockheed Martin Solar & Astrophysics Laboratory, 3. SETI Institute, 4. Niels Bohr Institute, University of Copenhagen

Understanding the coupled solar atmosphere—from the convection zone to the corona—requires simulations spanning multiple orders of magnitude in density, temperature, and scale.

While the legacy Bifrost code has driven fundamental advances, its architecture forces severe trade-offs between domain extent, resolution, and computational cost; resolving sharp chromospheric dynamics globally demands a prohibitively small timestep, while expanding the domain requires sacrificing local resolution. To remove these barriers, we introduce Dispatch-Bifrost, a ground-up reimplementation of the Bifrost physics suite on the Dispatch asynchronous, task-based framework. Dispatch-Bifrost eliminates traditional constraints through three key features:

Adaptive Mesh Refinement (AMR): Dynamically allocates resolution to follow the physics (e.g., current sheets and shock fronts) without wasting resources on smooth regions.

Local Timestepping: Replaces the global CFL constraint with localized, patch-specific timesteps, providing order-of-magnitude speedups that make routine deep-convection-to-corona runs practical.

Improved Radiative Transfer: Utilises a modern RT implementation designed for adaptive hierarchies, enabling accurate radiative coupling across the photosphere and chromosphere without legacy geometric constraints.

Furthermore, the framework supports multi-solver strategies that match numerical methods to local physical conditions within a single simulation.

We present early results from photosphere-to-corona test cases and performance benchmarks against legacy Bifrost, demonstrating its potential for high-fidelity solar atmospheric modelling.

Spectroscopic observations of global wave producing active regions

*David Long[1], Deb Baker[2], Andy Shu Ho To[3]

1. Dublin City University, 2. University College London, 3. ESA

Solar eruptions are often associated with global EUV waves, which can propagate at speeds of up to 1000 km/s through the low solar corona away from the eruptive active region. However, not all solar eruptions tend to produce these global waves, with the result that the coronal conditions and active region properties required to produce them remain poorly understood. Here we investigate ten global wave events and their origin active regions which were well observed by the Extreme ultraviolet Imaging Spectrometer (EIS) and Interface Region Imaging Spectrometer (IRIS) instruments. These very rare observations provide an opportunity to probe the evolution of mass and energy through the solar atmosphere before, during, and after the eruption of the global wave, enabling a better understanding of the physical processes which lead to these spectacular phenomena. These results will enable more targeted observations of eruptive active regions by future spectroscopic instruments such as MUSE and Solar-C.

SOLAR-C diagnostics of small-scale coronal events from simulated spectra

*Vincenzo Andretta[1], Fatemeh Z. Majidi[1], Costanza Argiroffi[2], Mariarita Murabito[3], Paolo Pagano[4], Fabio Reale[4], Clementina Sasso[1]

1. INAF/Capodimonte Astronomical Observatory, 2. INAF/Astronomical Observatory of Palermo, 3. INAF/Astronomical Observatory of Rome, 4. University of Palermo

One of the main objectives of the SOLAR-C mission is to understand how fundamental processes lead to the formation of the solar atmosphere and the solar wind. Within this overarching goal, a key science objective is investigating the contribution to coronal heating due to coronal events occurring at small scales. To achieve this objective, the SOLAR-C mission will utilize its EUV High-throughput Spectroscopic Telescope (EUVST) to perform seamless observations of all temperature regimes in the solar atmosphere from the chromosphere to the corona, up to 10 MK and beyond.
In this poster, we demonstrate the capability of EUVST to observe and resolve signatures that have so far been difficult or even impossible to observe. Examples include: small-scale reconnection events, in particular the so-called nanojets (small, bi-directional collimated plasma jets), and transverse waves as they emerge from the transition region and propagate into the corona. By utilizing detailed MHD simulations of such phenomena, we compute EUVST synthetic spectra convolved with its instrumental response. Guided by these simulations, we describe possible observing modes tailored to these science objectives. Finally, we briefly discuss perspectives for future observation with SOLAR-C and other upcoming or planned facilities/space missions.

Chromospheric and Coronal Heating at the Sites of Fine-Scale Magnetic Features (Observations from SO/EUI, SDO/AIA/HMI & Hinode/XRT)

*Kariyappa R.[1], Prabhu Ramkumar[2], Masuda S.[3], Zender J.[4], Dame L.[5]

1. Indian Institute of Astrophysics/ISEE, NU, 2. Kodaikanal Solar Observatory/Indian Institute of Astrophysics, 3. ISEE, NU, 4. ESA/ESTEC, 5. LATMOS/CNRS

The heating of the solar chromosphere and corona, the physical nature of the wave propagation, and the foot points of the chromospheric and coronal fine-scale structures with respect to the underlying photospheric magnetic features are extremely important issues in Solar Physics.
A 30-min time sequence images observed by Solar Orbiter/EUI on March 17, 2022 simultaneously in 174 Å(3s) & 1216 Å(12s) have been analysed. We selected totally 28 campfires/bps and derived the cumulative intensity values in the campfire/bps locations. The time series and the corresponding power spectrum plots have been generated. It is seen that the intensity variations in all the campfires/bps over the total observed period. We derived the period of intensity oscillations and noticed that the period is independent of their brightness variations. We observed a phase delay/shift of around 2-min between 174 Åand 1216 Åin campfires/bps time series, suggesting that the campfires are associated with the upward propagating waves. It looks from the intensity oscillations that the campfires are more diverse and dynamic in nature. Further studies are in progress (i) to bring out the differences among the campfires/bps, (ii) to bring out the differences and similarities between the different campfires and bright points, (iii) the phase delay/shift, (iv) heating mechanism in all the cases, and (v) estimate the energy contribution of campfires/bright points in the heating of the corona. The important results of the analysis will be presented in this poster paper.

Height Sensitivity of Milne-Eddington Atmospheres in Solar Inversions

*Jude Simmons[1], Andreas Korpi-Lagg[1], Alex Feller[1], Sami Khan Solanki[1], Damien Przybylski[1], Johann Hirzberger[1], Jan Staub[1]

1. Max Planck Institute for Solar System Research

The Photospheric Magnetic field Imager (PMI) is a space weather monitoring instrument onboard the ESA Vigil mission to the Lagrange point L5 that will measure the Zeeman-sensitive photospheric 617.3 nm spectral line. Through spectro-polarimetric observations, PMI will provide Doppler velocity and vector magnetograms at a 30-minute cadence. From this vantage point, magnetic structures will be detected 3-5 days earlier than from Earth, thus improving our ability to forecast eruptive regions.

In preparation for this mission, we simulate PMI-like observations and assess their performance. To this end, we compute synthetic Stokes profiles by solving the forward radiative transfer equation for a MURaM magnetohydrodynamics simulation. We then perform Milne–Eddington inversions across a range of representative spatial resolutions and viewing angles. The resulting height-independent inversion outputs are compared with slices of the simulation (in optical depth) to determine the height sensitivities of the respective PMI data products. This analysis provides an assessment of how instrument characteristics and heliocentric angle influence the inversion result. Response functions are used to verify the accuracy and reliability of the inversion solution, and may enable cross-calibration with other solar observatories. This work will guide the development of the PMI data reduction pipeline and refine its data products.

fiasco: A Python interface to the CHIANTI atomic database

*Will Barnes[1,2], Jeffrey Reep[3], Peter Young[2], Nabil Freij[4], Laura Hayes[5], Stuart Mumford[6], Nick Murphy[7], Jacob Parker[8], David Stansby[9]

1. American University, 2. NASA Goodard Space Flight Center, 3. University of Hawai'i at Manoa, 4. SETI Institute, 5. Dublin Institute for Advanced Studies, 6. Aperio Software, 7. Harvard-Smithsonian Center for Astrophysics, 8. Montana State University, 9. University College London

The CHIANTI atomic database is a critical tool for understanding remote-sensing observations from across the electromagnetic spectrum. The database consists of energy levels, transition wavelengths, excitation and decay rates, and ionization and recombination rates for nearly 500 ions from 30 different elements. The accompanying software for accessing these data and computing derived quantities like the contribution function or ionization balance has traditionally been developed in the Interactive Data Language (IDL) and distributed through SolarSoft. The fiasco package is a modern Python interface to the CHIANTI atomic database and provides much of the same functionality as the existing IDL tools through an object-oriented interface. In this poster, we will describe the design of the fiasco package, discuss the current state and future direction of the software, and show how to perform some common analysis tasks with fiasco. Additionally, we will show current benchmarks against the existing IDL tools as well as some examples of how fiasco is being used for scientific analysis.

Bifrost spectral inversions. Fast non-LTE solar chromospheric diagnostics from 3D simulations.

*Tiago M. D. Pereira[1,2], Elias R. Udnæs[1,2], Ignasi J. Soler Poquet[1,2], Luc Rouppe van der Voort[1,2]

1. Rosseland Centre for Solar Physics, University of Oslo, 2. Institute of Theoretical Astrophysics, University of Oslo

Modern solar observatories produce vast amounts of detailed spectral data. To infer atmospheric parameters from these data, in particular for the solar chromosphere, requires immense computational resources. This problem is becoming more acute as spatial resolution improves, and faster methods are much needed to make sense of the data. Archive-based inversions provide a cost-effective way to infer atmospheric parameters from solar spectra, and involve building databases of synthetic spectra and atmospheric models. We extend this idea to a *k*-nearest neighbour regression with non-LTE chromospheric spectra from a three-dimensional radiative-magnetohydrodynamic simulation. Our database consists of 150 million Ca II 854.2 nm line profiles from a magnetically quiet simulation. We apply the database inversion on high-cadence observations taken with the Swedish 1-m Solar Telescope and find excellent agreement between the observed and fitted spectra for almost the entire field of view. We recover the chromospheric temperature, gas density, and line-of-sight velocity, and calculate the uncertainty in their inversions. Temperature and line-of-sight velocities have small uncertainties over a broad range of optical depths, and are reliably recovered over a larger optical-depth range than with a traditional inversion method. Compared with the traditional inversion method, our approach is four orders of magnitude faster. Limitations and future improvements of the method are discussed.

XRTDEMIterative: Open-Source Python DEM Analysis for Hinode/XRT in xrtpy

*Katharine Reeves[1], Joy Velasquez[1], Mark Weber[1], Nicholas Murphy[1]

1. Harvard-Smithsonian Center for Astrophysics

The Differential Emission Measure (DEM) describes how plasma is distributed along the line of sight as a function of temperature in the solar corona, and deriving it from multi-filter X-ray observations is essential to studying coronal heating, flare energetics, and the thermal evolution of solar phenomena. Hinode/XRT - currently the only operating solar soft X-ray telescope, with nine filters spanning a broad range of coronal temperatures - offers a uniquely powerful window into this thermal structure, yet this analysis has historically relied on the IDL routine xrt_dem_iterative2.pro, leaving a gap for the growing portion of the heliophysics community working entirely in Python. We present XRTDEMIterative, a Python implementation of this solver fully integrated into xrtpy - the open-source Python package for Hinode/XRT data analysis.The algorithm parameterizes log10(DEM) as a cubic spline across a user-defined log10(T) grid and iteratively minimizes residuals between observed XRT filter intensities and a forward model computed by integrating the DEM against each filter's temperature response, with Monte Carlo uncertainty propagation achieved by solving an ensemble of Gaussian-perturbed realizations of the input intensities. XRTDEMIterative is delivered as a complete, tested, and documented package —including a full unit test suite, API documentation, built-in plotting tools, and worked Jupyter notebook examples covering end-to-end workflows from raw XRT intensities to DEM solutions with Monte Carlo uncertainties —natively integrated with xrtpy's temperature response infrastructure and requiring no IDL or SSW dependency, making reproducible XRT thermal analysis fully accessible to the modern heliophysics community.

Development of a divergence-free and well-balanced scheme for the magnetohydrodynamic relaxation method

*Takahiro Miyoshi[1], Satoshi Inoue[2], Daiki Yamasaki[3], Shin Toriumi[3], Kanya Kusano[4]

1. Hiroshima Univ., 2. NJIT, 3. JAXA / ISAS, 4. Nagoya Univ.

Various phenomena in the solar atmosphere are driven by the solar magnetic field. Since direct measurements of the coronal magnetic field with high accuracy remain difficult, numerical methods for extrapolating the three-dimensional magnetic field from photospheric observations have been extensively developed. The magnetohydrodynamic (MHD) relaxation method is a powerful approach in which equilibrium solutions are obtained as converged states of simplified MHD equations. In particular, we previously developed a robust and divergence-free MHD relaxation scheme for nonlinear force-free magnetic field modeling by combining an upwind-type numerical scheme with a constrained-transport (CT) method. Here, we extend the scheme to magnetohydrostatic (MHS) magnetic field modeling including gas pressure and gravity.
In the present scheme, the magnetic field and pressure are evolved separately. The magnetic field evolution is solved using the HLLD approximate Riemann solver, while the pressure evolution is computed with the Lax–Friedrichs method. An upwind CT scheme is employed to preserve the solenoidal condition of the magnetic field. Furthermore, to maintain hydrostatic equilibrium numerically, the gravitational source term is evaluated using cell-interface values obtained after subtracting the local hydrostatic pressure distribution within each computational cell.
Numerical experiments demonstrate that the proposed scheme robustly reconstructs stable MHS equilibrium solutions while maintaining both divergence-free and well-balanced properties.

Full-disk distributions of photospheric and chromospheric oscillations

*Haruhi Shirato[1], Kiyoshi Ichimoto[1], Satoru UeNo[1]

1. Kyoto University

Understanding the relationship between waves and magnetic fields is essential for revealing the mechanisms of atmospheric heating. In this study, we analyzed chromospheric velocity data lasting more than 12 hours in the H-alpha line core obtained with SMART/SDDI, together with photospheric velocity data from SDO/HMI. We examined the spatial distributions of oscillation power and phase travel time over the entire solar disk. We detected 5-minute oscillations in the photosphere and both 3-minute and 5-minute oscillations in the chromosphere. In chromospheric power maps, oscillation power in both period bands is suppressed in global enhanced magnetic regions. Since no corresponding suppression is seen in the photospheric power, and the phase travel time is nearly zero in these regions. These results may suggest that cut-off frequency is higher, or that propagating waves are reflected before reaching the chromosphere.In center-to-limb analysis of quiet-Sun regions, for 5-minute oscillations, the amplitudes of photospheric and chromospheric oscillations decrease monotonically toward the solar limb, while the amplitude ratio of chromosphere to the photosphere and the phase travel time increase monotonically. For 3-minute oscillations, the amplitudes of photospheric and chromospheric oscillations decrease monotonically toward the solar limb, while the amplitude ratio and the phase travel time between the two layers remain nearly constant regardless of the distance from the disk center.Interpreted as slow magnetoacoustic waves, both the 3-minute and 5-minute oscillations propagate along magnetic field lines; however, the propagation of the 5-minute oscillations depends on the field inclination, whereas that of the 3-minute oscillations does not.We discuss these findings in relation to magnetic field structure.

Microwave Spectral Observations of Solar Flares Using the Yokosuka Radio Polarimeter

*Kogaku Kitayama[1], Kyoko Watanabe[1], Ryosuke Okubo[1], Kazumasa Iwai[2], Satoshi Masuda[2], Masumi Shimojo[3]

1. National Defense Academy of Japan, 2. Institute for Space-Earth Environmental Research, Nagoya University, 3. National Astronomical Observatory of Japan

The microwave spectrum from solar flare exhibits a spectral shape with a peak at the turn-over frequency, resulting from the superposition of radiation from optically thick and thin regions. It is known that this turn-over frequency depends on the magnetic field strength and plasma electron density in the flare region (Dalk 1985). The spectral slope (power-law) on the high-frequency side serves as an important indicator reflecting the energy distribution of non-thermal electrons accelerated during the flare. Although spectra from solar flares were also obtained by the Nobeyama Radio Polarimeters (NoRP), the observation frequencies were limited; therefore, the spectra during flares were derived by interpolating and extrapolating the observed data. In contrast, the Yokosuka Radio Polarimeter (YoRP), developed as the successor to NoRP, achieves automated solar tracking and routine observations of the solar microwave in the 2.8–10 GHz band with high frequency resolution (0.0075 GHz) and high temporal resolution (approximately 7 seconds), making it possible to obtain detailed solar flare microwave spectra and their temporal variations.

Since the start of routine observations in February 2026, YoRP has observed some solar flare events. In this presentation, we report the analysis results of three selected cases from multiple flare spectra observed by YoRP observations. In the X1.5 flare on March 30, 2026, the existence of non-thermal accelerated electrons in a large flare were confirmed. In the M7.5 flare on April 4, microwave emission was observed to persist even after the X-ray peak, indicating continuous electron acceleration above the flare loops. In the M1.7 flare on April 23, short-duration non-thermal emission was observed, confirming the typical behavior of an impulsive flare.

In the future, through the routine operation of YoRP, we aim to collaborate with the SOLAR-C project to provide data that contributes to space weather forecasting.

Magnetic Portals in Quiet-Sun Flux Tubes Revealed by Chromospheric Spectropolarimetry with SUNRISE-3/SCIP

*Takayoshi Oba[1,2], Yukio Katsukawa[3], Masahito Kubo[3], Yusuke Kawabata[3], Takuma Matsumoto[4], Ryohtaroh Ishikawa[5], Yoshihiro Naito[6], Toshifumi Shimizu[7], Hirohisa Hara[3], Quintero Noda Carlos[9], Jose Carlos del Toro Iniesta[8], David Orozco Suárez[8], Sami Solanki[2], Andreass Lagg[2], Achim Gandorfer[2], Pietro Bernasconi[11], Thomas Berkefeld[10], Alex Feller[2], Tino Riethmüller[2], Sunrise III Team

1. Advanced Research Center for Space Science and Technology (ARC-SAT), Institute of Science and Engineering, Kanazawa University, 2. Max-Planck-Institut für Sonnensystemforschung, 3. National Astronomical Observatory of Japan, 4. Centre for Integrated Data Science, Institute for Space-Earth Environmental Research, Nagoya University, 5. National Institute for Fusion Science, 6. The Graduate University for Advanced Studies (SOKENDAI), 7. Institute of Space and Astronautical Science, Japan Aerospace Exploration Agency, 8. Instituto de Astrofísica de Andalucía, 9. Instituto de Astrofísica de Canarias, 10. Institut für Sonnenphysik (KIS), 11. Johns Hopkins University Applied Physics Laboratory

Acoustic waves propagate into the chromosphere and upper atmosphere, contributing to energy transport. Their upward propagation is restricted to frequencies above the *acoustic cutoff frequency*, depending on atmospheric conditions. The magnetic field configuration plays a key role, as the cutoff frequency is reduced in regions where the field is inclined with respect to gravity, forming so-called *magnetic portals*. Previous studies linked magnetic fields and oscillations in quiet regions, but these analyses were based on photospheric magnetic field information, leaving chromospheric structure unconstrained. This study investigates the coupling between acoustic waves and magnetic topology using photospheric and, for the first time, chromospheric spectropolarimetry in a quiet region, obtained with the Sunrise Chromospheric Infrared SpectroPolarimeter (SCIP) aboard the Sunrise III balloon mission in 2024. The SCIP slit samples magnetic features in which the line-of-sight field strength exhibits a sharp spatial peak in the photosphere while becoming broader and weaker in the chromosphere, indicating expanding fluxtubes. In chromospheric velocity field, these fluxtubes exhibit strong 5-minute oscillations, while the surrounding regions show weak 3-minute oscillations. The enhanced oscillations occur in fluxtubes that not only expand but also show offsets between the photospheric and chromospheric magnetic centroids, indicative of possibly tilted tube axes. In these regions, sawtooth temporal velocity variations are associated with intensity enhancement, suggesting steepened shocks. Fluxtubes with low-frequency oscillations are identified in network regions and, thanks to the performance of SCIP and Sunrise, are also discernible in internetwork regions, barely detectable in SDO/HMI magnetograms. These results provide observational evidence that expanding fluxtubes, possibly with tilted axes, act as magnetic portals in network and internetwork regions, allowing low-frequency waves upward and driving chromospheric dynamics via shocks.

10 years of solar observations with ALMA

*Masumi Shimojo[1]

1. National Astronomical Observatory of Japan

The solar observing mode have been offered as one of the observing modes of ALMA to the community since Cycle 4 (October 2016 –September 2017). This year marks the 10th anniversary of ALMA solar observations. ALMA solar observations began with interferometric observations using two receiver bands (Band 3 [100 GHz] and Band 6 [239 GHz]) and full-sun mapping using the TP array. New solar observing modes have been developed over the past decade, and we can now use the modes: Band5 [198 GHz] and Band7 [346 GHz] receivers, fast regional mapping with the TP array and polarization observations with Band3.

Over the past decade, over 50 ALMA solar observing projects have been carried out, and these mm-wave solar datasets have produced new knowledge about solar structures and phenomena. Some of these finding are summarized in the special issue 'The Sun Seen with the Atacama Large Millimetre/submillimetre Array (ALMA): First Results', published by Frontiers in 2023.

In this presentation, we summarize the development history of ALMA solar observation modes and the knowledge gained from mm-wave solar observations. Furthermore, we will discuss the future of ALMA solar observations, particularly in the era following the Wide-band Sensitivity Upgrade (WSU), also known as "ALMA2" in Japan, which is a renewal project currently being carried out by the ALMA observatory.

Magnetic Field associated with an Active Region Filament Observed by SUNRISE III/SCIP in the Ca II 8542 Å Line

*Takuma Matsumoto[1], Yukio Katsukawa[2], Masahito Kubo[2], Yusuke Kawabata[2], Takayoshi Oba[4], Ryohtaroh T Ishikawa[3], Yoshihiro Naito[5], Toshifumi Shimizu[6], Hirohisa Hara[2], Toshihiro Tsuzuki[2], Kazuya Shinoda[2], Tomonori Tamura[2], Sami K Solanki[8], Andreas Lagg[8], Achim Gandorfer[8], Pietro Bernasconi[10], Thomas Berkefeld[9], Alex Feller[8], Tino L Riethmüller[8], Jose Carlos del Toro Iniesta[7]

1. Nagoya University, 2. NAOJ, 3. NINS, 4. Kanazawa University, 5. SOKENDAI, 6. ISAS/JAXA, 7. IAA, 8. MPS, 9. KIS, 10. APL

We present high-spatial-resolution spectropolarimetric observations of a solar filament, captured on 15 July 2024 by the Sunrise Chromospheric Infrared spectro-Polarimeter (SCIP) during the Sunrise III balloon-borne mission. The target filament, situated near the solar disk center and adjacent to an active region, remained in a quiescent state throughout the two-hour observation window.Utilizing full Stokes profiles of the Ca II 8542 Å line, SCIP successfully detected distinct linear polarization signatures. These signals, which exceeded the 2-sigma noise level, exhibited the characteristic double-lobe spectral profile of the transverse Zeeman effect, clearly distinguishing them from scattering-induced polarization.Preliminary analysis using the weak field approximation indicates a magnetic field strength of approximately -80 G along the line of sight, with the transverse component ranging between 300 and 500 G. These measurements likely characterize the magnetic properties of both the filament and its supporting chromospheric structure. While the magnetic field vector aligns nearly parallel to the filament axis in the northeastern section, the southeastern portion extended beyond the instrument's field of view.To our knowledge, this study represents the first unambiguous detection of linear polarization associated with a solar filament using the Ca II 8542 Å line. These findings establish a new diagnostic framework for investigating the vector magnetic field of filaments in the lower chromosphere, providing a vital complement to existing He I diagnostics that sample the upper chromosphere.

Two-Zone Fokker-Planck Approach for Modeling the Time Evolution of Non-thermal Emissions in Solar Flares

*Shunsaku Nagasawa[1], Dmitry Khangulyan[2], Tadayuki Takahashi[1,3]

1. Kavli IPMU, The University of Tokyo, 2. Institute of High Energy Physics, Chinese Academy of Sciences, 3. QUP, KEK

The time evolution of non-thermal emissions in solar flares provides critical insights into the underlying particle acceleration and transport mechanisms. In particular, a Soft-Hard-Soft (SHS) spectral variation in hard X-rays, where the power-law index hardens as the flux increases toward the peak, then softens as the flux decreases, has been frequently observed in solar flares. This behavior is believed to reflect underlying particle acceleration or energy losses, but the detailed description of these processes is still missing. During the impulsive phase of the M7.6-class flare on July 23, 2016, we identified a characteristic pattern seen in the plane X-ray flux-spectral index. The spectrum is harder during the rising phase than during the decay phase characterized by the same flux level, thus with time the flare parameter follow a clockwise trajectory in this plane.
In this study, we develop a time-dependent two-zone model to investigate the mechanisms responsible for this spectral evolution. The model consists of two distinct regions: zone-1, representing the flare loop-top where non-thermal electrons are injected and undergo energy losses; and zone-2, representing the flare footpoints where electrons accumulate after escaping from zone-1 and emit bremsstrahlung photons. Using numerical approach based on the Fokker-Planck equation, we perform a systematic analysis of the time evolution of electron distributions and synthetic hard X-ray spectra for various loss and escape conditions. While the two-zone model captures key aspects of electron transport and cooling, we find that it alone cannot reproduce the observed clockwise pattern under the assumption of fixed injected spectrum of accelerated electrons. This suggests that dynamic changes in the properties of the injected electron population, specifically simultaneous variations in both spectral slope and flux, are crucial for reproducing the observed clockwise behavior, which is likely a signature of changes in the acceleration process itself rather than solely due to cooling and transport effects.

The ionospheric parameter related to forecasts of HF blackouts

*Kyoko Watanabe[1], Shinnosuke Kitajima[1], Hidekatsu Jin[2], Chihiro Tao[2], Satoshi Masuda[3], Michi Nishioka[2]

1. National Defense Academy of Japan, 2. National Institute of Information and Communications Technology, 3. Nagoya Univ.

Shortwave fadeouts (SWF) are sudden disruptions of high-frequency (HF) radio communication caused by enhanced solar X-ray emission during flares. Traditionally, the magnitude of SWF has been estimated empirically based on the peak solar X-ray flux and the solar zenith angle. However, this flare-centric approach often fails to accurately predict the severity of ionospheric absorption, particularly when distinguishing between simple absorption enhancements and complete blackouts. Recent observations have revealed that ionospheric conditions before flare critically modulate the impact of flares, and this effect appears to be particularly pronounced during ionospheric negative storms.We performed a comprehensive statistical analysis using a long-term dataset spanning over 40 years (1981–2023) obtained from four ionosonde stations in Japan. From this analysis, we derived a new ionospheric susceptibility parameter, "fB," defined as the ratio of the background F2-layer critical frequency (foF2) to the difference between foF2 and the minimum frequency (fmin).The fB exhibits distinct seasonal and diurnal variations independent of the solar cycle, quantitatively representing the background ionosphere's vulnerability to blackouts. Furthermore, we found that a high fB value significantly lowers the flare intensity threshold required to trigger a blackout. By incorporating fB into a new empirical model, the accuracy of blackout prediction improved significantly, with the Area Under the Curve (AUC) increasing from 0.75 to 0.82 compared to conventional methods. These findings demonstrate that monitoring fB enables real-time, pre-flare risk assessment, providing a robust foundation for improving operational space weather forecasting. Furthermore, information on the flare EUV spectrum obtained by SOLAR-C will make it possible to alert HF blackout in the future.

The Hα Line as a Probe of Chromospheric Magnetic Fields

*Harsh Mathur[1,2], Jayant Joshi[2], Thore Espedal Moe[1], Tiago Pereira[1], Nagaraju K[2]

1. Rosseland Centre for Solar Physics / University of Oslo, 2. Indian Institute of Astrophysics

The solar chromosphere constitutes a key interface governing the transfer of mass and non-thermal energy from the lower atmosphere into the corona, yet its magnetic structure remains insufficiently constrained due to limitations of existing spectropolarimetric diagnostics. Although the Hα line is among the most extensively observed chromospheric diagnostics, its suitability for magnetic field measurements has historically been questioned based on conclusions drawn from simplified one-dimensional radiative-transfer models suggesting dominant photospheric contributions to polarization signals. Recent observational studies (Mathur et al. 2023, 2024) indicate that the Hα line core samples magnetic fields at significantly higher chromospheric layers than commonly used diagnostics such as Ca II 8542. Here, we revisit the magnetic sensitivity of Hα using full three-dimensional non-LTE radiative transfer synthesis performed in realistic radiative-MHD simulations (Mathur et al. 2025). Our results demonstrate that the Hα line core forms substantially higher in the chromosphere, with peak magnetic response occurring near log τ500 ≈−5.7, enabling direct probing of upper-chromospheric magnetic fields. Synthetic Stokes profiles further show that measurable Stokes V signals persist under realistic polarimetric noise levels (˜1e-3 Ic), tracing line-of-sight magnetic fields within highly structured fibrilar atmospheres. In this contribution, I discuss these modeling results and the expected Hα polarization signal amplitudes under realistic noise conditions representative of next-generation high-sensitivity facilities such as the European Solar Telescope and DKIST, highlighting the diagnostic potential of Hα spectropolarimetry for constraining magnetic structure and energy transport in the upper chromosphere.

Spectropolarimetric observations of coronal oscillations with DKIST/CryoNIRSP

*Petra Kohutova[1]

1. Rosseland Centre for Solar Physics, University of Oslo

The CryoNIRSP instrument at Daniel K. Inouye Solar Telescope provides spectropolarimetric observations of the solar corona in near-infrared wavelengths with spatial resolution and temporal cadence unprecedented for coronal polarimetry. We present observations of oscillating coronal loops obtained during DKIST Observing Cycle 3 using the off-limb coronal mode in the Fe XIII 1074 and 1079 nm coronal emission lines. Doppler velocity measurements combined with imaging data were used for oscillation detection and analysis, and full Stokes spectra served as diagnostics of coronal magnetic field properties. Together, these complementary diagnostics provide new insights into properties and damping mechanisms of coronal oscillations, as well as into the evolution of the large- and fine-scale structure of the solar corona.

Full-Disk Spectroscopy of the Solar Corona Across a Solar Cycle with Hinode/EIS

*James McKevitt[1,2], Ignacio Ugarte-Urra[3], Peter Young[4,5]

1. Mullard Space Science Laboratory, University College London, 2. Institute of Astrophysics, University of Vienna, 3. Space Science Division, Naval Research Laboratory, 4. Solar Physics Laboratory, Heliophysics Science Division, NASA Goddard Space Flight Center, 5. Department of Mathematics, Physics and Electrical Engineering, Northumbria University

The structure and dynamics of the solar corona evolve with the Sun's magnetic cycle, yet how this variability manifests in the disk-integrated, Sun-as-a-star observables used in stellar activity studies remains poorly constrained. We compile 18 full-disk spectroscopic mosaic scans made by Hinode/EIS spanning 2013-2024, covering solar cycle 24 and the rise of solar cycle 25, and probe coronal plasma variability through the integrated and spatially resolved intensity, Doppler velocity, and non-thermal velocity of logT˜6.2 plasma in active regions and the quiet Sun. Disk-integrated coronal intensity is strongly correlated with the solar cycle, consistent with stellar observations. No clear solar-cycle variation is found in the distributions of coronal Doppler or non-thermal velocity in either active regions or the quiet Sun, though their total intensities do track the cycle. Active region intensity per unit solid angle shows a moderate correlation with solar cycle. Taken together, these results support the hypothesis that Sun-as-a-star coronal intensity variability across the solar cycle is driven primarily by the changing fraction of the disk occupied by active regions, rather than by changes in the logT˜6.2 plasma properties of those regions. The well-established correlation between upflowing plasma and elevated non-thermal line broadening in active regions persists throughout the cycle, implying that the underlying kinematic properties of active region plasma are insensitive to the global magnetic field configuration, a result with direct implications for the interpretation of coronal activity cycles on solar-like stars.

Sun-as-a-star Research: Response of Chromospheric Lines to Sunspot Area and Complexity

*Junya Natsume[1], Ayumi Asai[1], Satoru Ueno[1], Takato Otsu[2], Kai Ikuta[3], Takako Ishii[1], Sanghee Lee[1], Kazunari Shibata[4,1]

1. Kyoto University, 2. JAXA/ISAS, 3. Hitotsubashi University, 4. Doshisha University

Sunspots are the energy source of solar flares, and their area and complexity are strongly related with both flare magnitude and occurrence frequency. Superflares–flares with energies more than an order of magnitude greater than the largest solar flares–have been observed on stars, and they are often accompanied by giant starspots. Like solar flares, the occurrence frequency of stellar flares is thought to be related to the complexity of starspots. However, due to spatially unresolved observations, the complexity of starspots is not well understood, making it more difficult to evaluate stellar activity than that of the Sun. In this study, we investigated how sunspot complexity manifests itself in chromospheric lines, based on a "Sun-as-a-star" analysis, in which solar data are spatially integrated to mimic stellar observations. While previous studies have examined the relationship between the solar magnetic flux and the Sun-as-a-star irradiance of solar spectral lines, there have been very few studies focusing on sunspot complexity. We conducted imaging-spectroscopic observations of solar active regions with various levels of complexity (α-, β-, γ-, and δ-type) in the Hα and Ca II K lines using Domeless Solar Telescope at Hida Observatory, Kyoto University. For 50 active regions with different types, we measured the sunspot areas and applied a Sun-as-a-star analysis with these lines to derive equivalent widths representing the difference in brightness relative to quiet regions. We found that the equivalent widths of active regions tend to increase with increasing sunspot area for both Hα and Ca II K lines, and the equivalent widths of δ-type spots are larger than those of α-type spots, as the former generally has a larger area than the latter. In this presentation, we discuss the details of these results, the chromospheric structures that control the magnitude of the equivalent widths, and the implications for starspots.

Hinode SOT/SP and SDO/HMI Observations of the Polar Magnetic Field (2010-2026)

*Nariaki Nitta[1]

1. Lockheed Martin Advanced Technology Center

Hinode SOT/SP observations of the Sun's polar regions have revealed small patches reminiscent of polar faculae. These patches are characterized by enhanced magnetic fields in the kilogauss range. Two HOPs are dedicated to observing these polar fields. While analyzing the normal (full-resolution) maps from HOP 81, we utilized SDO/HMI data to correct the pointing. During this process, we noted that HMI full-disk vector magnetograms contain similar magnetic patches in the polar regions. Furthermore, the average polar magnetic fluxes calculated from SP and HMI data show a striking resemblance over the solar cycle. However, their absolute fluxes do not match; HMI fluxes tend to be several times higher. This discrepancy may be partly attributed to the limitations of HMI data regarding spatial resolution and Stokes polarimetry. Conversely, previous studies have pointed out that SP data may not provide the absolute ground truth, largely due to its own spatial resolution limits. In this work, we extend the comparison of individual patches observed by both SP and HMI during the 2010–2026 period. We carefully select patches from HMI data at a 12-minute cadence that coincide with nearly simultaneous SP scans, which require approximately 7.5 hours to complete. This study aims to determine whether the differences in magnetic flux between SP and HMI data depend on the physical properties of the patches and the phase of the solar cycle.

Noise Estimation for Solar Polar Missions: the Case of Travel Time Measurements to Infer the Solar Meridional Circulation

*Yoshiki Hatta[1], Hideyuki Hotta[1], Takashi Sekii[2,3], Yuki Takao[4]

1. Institute for Space-Earth Environmental Research / Nagoya University, 2. National Astronomical Observatory of Japan, 3. SOKENDAI, 4. Yokohama National University

In the time-distance helioseismology, we measure the time it takes waves to propagate between two points on the solar surface, based on which we can infer the structure and dynamics around the wave propagation region. Nevertheless, we often suffer from noise that is caused by stochasticity in the turbulent convection. It is especially the case when we use Dopplergrams to measure travel times in high-latitude regions. This is because, in the Doppler velocity, the horizontal component of the surface velocity field (to which convection contributes more than oscillation) becomes more dominant over the radial component of the velocity field (to which oscillation contributes more than convection) in higher-latitude regions. Such effects of line-of-sight projection can be partly alleviated by, e.g., carrying out two-vantage-point observation of the Sun, namely, one on the ecliptic plane and the other from high ecliptic latitudes. In this talk, focusing on travel time measurements to infer the solar meridional circulation, we would like to show you to what extent the noise in travel times would be reduced by conducting the two-vantage-point observation of the Sun. To this end, we take a semi-analytical approach basically following the analytical study by Gizon & Birch (2004). We found that the variance (of the travel time) becomes significantly smaller for high-latitude regions if we have another observation from high ecliptic latitudes. It is, nevertheless, also found that the minimum value of the variance is not so changed from that obtained by single-vantage-point observations. Since there has never been a quantitative evaluation of noise that would result from the two-vantage-point observation of the Sun, our results may be quite helpful for feasibility tests of future solar polar missions.

The Short-wavelength Camera for SOLAR-C EUVST

*Sarah Matthews[1], Duncan Rust[1], Berend Winter[1], James McKevitt[1], Saeeda Awan[1], Diagarajen Carpanen,[1], Tom Kennedy[1], Ted Pyne[1], Nuwaid Nattukallingal[1], Anand Pendem[1], Sampie Smit[1], Dave Walton[1], Ian Phillips[1], Abdullah Khalil[1], Anamarija Stankov[3], Mark Hailey[1], Ryohko Ishikawa [2], Fumihiro Uraguchi[2], Yukio Katsukawa[2], Hiro Hara[2]

1. UCL Mullard Space Science Lab., 2. NAOJ, 3. ESA

The EUVST spectrograph (EUV High-throughput Spectroscopic Telescope) onboard the SOLAR-C mission (JAXA) is designed to simultaneously observe the solar atmosphere from the 10,000-Kelvin chromosphere to the flaring corona (> 10 million-Kelvin) at 0.4 arcsecond spatial resolution, allowing the continuous tracking of plasma and energy transport throughout the atmosphere. The spectrograph consists of one short wavelength (SW) CCD camera (under development by ESA/UCL-MSSL), three long wavelength (LW) CMOS cameras and a slit-jaw imager (under development by NASA/NRL/LMSAL). We describe the design and status of the SW camera development, including initial science performance expectations.

Synthetic chromospheric nanoflare observations using RADYN

*Aline Rangøy Brunvoll[1], Sven Wedemeyer[1], Mats Carlsson[1], Mats Kirkaune[1]

1. Rosseland Centre for Solar Physics, Institute of Theoretical Astrophysics, University of Oslo

Solar nanoflares are theorized to be a large contributor to the heating of the solar atmosphere. But as they are thought to produce very small and weak emission signatures compared to the background of the sun, it makes them difficult to observe with confidence.
Here we present a selection of synthesized nanoflare emission signatures in the chromospheric spectral lines. The result gives us an idea of what nanoflare spectra would look like in observations of the solar chromosphere, and can aid us in observing them with e.g. the Swedish Solar Telescope (SST).
Our RADYN 1D nanoflare simulations are based on the pre-existing F-CHROMA grid of flare simulations, which we initialized with lower energies than the original grid. We synthesized the chromospheric lines using the RH 1.5D code, and studied their formation depending on the parameters of the models.
We conclude that the analysis of chromospheric lines such as Ca II K can be useful in identifying nanoflares and separating them from other events with similar signatures, like shockwaves. In practice, however, this would require observations taken with very high temporal cadence and spatial and spectral resolution. Organizing co-observations by complementary telescopes and instruments may be critical in order to identify nanoflares in observations with certainty.

Three-dimensional Magnetic Field Structure of a Quiet Sun Region as Revealed by Sunrise III/SCIP

*Masahito Kubo[1], Yukio Katsukawa[1], Yusuke Kawabata[1], Takayoshi Oba[2], Hirohisa Hara[1], Toshifumi Shimizu[3], Ryohtaroh T Ishikawa[4], Takuma Matsumoto[5], Yoshihiro Naito[6], Jose Carlos del Toro Iniesta[7], David Orozco Suarez[7], Carlos Quintero Noda[8], Sami K Solanki[9], Andreas Lagg[9], Achim Gandorfer[9], Pietro Bernasconi[10], Thomas Berkefeld[11], Alex Feller[9], Tino L Riethmüller[9], Alberto Alvarez-Herrero[12]

1. National Astronomical Observatory of Japan, 2. Kanazawa University, 3. ISAS/JAXA, 4. National Institute for Fusion Science, 5. Nagoya University, 6. Graduate University for Advanced Studies, Sokendai, 7. Instituto de Astrofísica de Andalucía, 8. Instituto de Astrofísica de Canarias, 9. Max Planck Institute for Solar System Research, 10. Johns Hopkins University Applied Physics Laboratory, 11. Institut für Sonnenphysik, 12. Instituto Nacional de Técnica Aerospacial

The Sunrise Chromospheric Infrared spectroPolarimeter (SCIP) onboard the Sunrise III balloon-borne stratospheric solar observatory obtained an unprecedented dataset for a quiet sun region near the disk center on July 11, 2024. SCIP is a slit-scanning spectropolarimeter that observes the polarization state of many spectral lines simultaneously in the 850 nm and 770 nm bands, covering both photospheric and chromospheric spectral lines. The full field of view of 58" x 58" was scanned in 107 minutes without interruption. With an integration time of 10 s per slit position, a remarkably stable polarimetric precision of 0.03–0.04% (1-sigma) of the continuum level was achieved. The multi-wavelength SCIP observations reveal that the chromospheric line-of-sight magnetic field exhibits thread-like elongated structures over the internetwork regions of the quiet Sun. No clear photospheric counterpart is detected directly beneath these chromospheric thread-like structures. The threads typically have widths smaller than 1” and are embedded within the canopy fields extending from nearby network regions. Their line-of-sight field strengths derived from the weak-field approximation are typically 10–20 G weaker than the surrounding canopy fields. In particularly clear cases, the magnetic polarity of thread-like structures is opposite to that of the adjacent canopy field. These findings suggest that the canopy field is not a smooth, monotonically expanding magnetic fan emanating from network regions but instead has a complex three-dimensional configuration containing numerous local dip structures. These observations provide new constraints on the three-dimensional magnetic topology of the photosphere and chromosphere in quiet-sun regions.

North pole spectropolarimetric observation by Hinode and Solar Orbiter

*Daniele Calchetti[1], David Orozco Suarez[2], Julian Blanco Rodriguez[3], Alejandro Moreno Vacas[2], Gherardo Valori[1], Johann Hirzberger[1], Sami Solanki[1]

1. MPS, 2. IAA, 3. Valencia Univ.

Direct remote-sensing observations of the solar poles and polar faculae have always been hindered by the limited viewing geometry achievable from Earth. Nevertheless, Hinode has monitored for many years the evolution of the polar magnetic field thorugh the solar cycle with high sensitivity and resolution. Since February 2025, Solar Orbiter started observing the Sun from out of the ecliptic plane, giving an unprecedented view of the poles.
In this work we compare the spectropolarimetric data obtained with SO/PHI-HRT and Hinode SOT/SP of the solar north pole in October 2025. The observations were performed while Solar Orbiter was near the Sun-Earth line, but at almost 17 degrees in heliographic latitude, whereas Hinode observed the same pole from an Earth perspective.
We discuss the similarities and differences between the two viewpoints and assess their implications for the retrieval of the polar magnetic flux and the structure of the magnetic canopy.

Spectral variability of the Ca II K line core in an emerging flux region from high-resolution Sunrise iii/SUSI observations

*Kunal Hardeep Singh[1], Alex Feller[1], Francisco A. Iglesias[2], Sami K. Solanki[1], Andreas Lagg[1], Achim Gandorfer[1], Juan Sebastián Castellanos Durán[1], Johannes Hölken[1], Edvarda Harnes[1], Tino L. Riethmüller[1]

1. Max Planck Institute for Solar System Research, 2. Grupo de Estudios en Heliofísica de Mendoza

Small-scale impulsive energy events, often attributed to magnetic reconnection, are candidates for contributing to the heating of the solar corona. However, the direct detection of the effect of such events in the corona remains challenging due to fast energy dissipation in the corona and the limited sensitivity of EUV/X-ray observations. These events, although faint in the corona, can drive significant chromospheric dynamics, leaving detectable imprints in strong resonance lines such as the Ca II K line, which forms over a broad height range from the lower photosphere to the higher chromosphere. In this study, we analyse high-resolution, full-Stokes spectropolarimetric observations from the Sunrise III/SUSI instrument, capturing an explosive event within an emerging flux region (NOAA AR 13738) on July 10, 2024. Using spectral moments, we characterise the variability of the Ca II K line core, focusing on regions exhibiting extreme line asymmetries and broadening. By comparing these moments with simultaneous photospheric and lower chromospheric observations from SUSI and SDO/AIA, we establish a multi-height context for the event. Our analysis reveals a localized, transient brightening in the AIA 17.1 nm and 9.4 nm channels, indicative of plasma temperatures reaching 1–7 MK.These coronal signatures are coincident with a strong, spatially confined enhancement in the Ca II K zeroth, first and second order spectral moments. Notably, the event is also spatially coincident with mixed-polarity magnetic patches and shows a loop-like connection to the boundary of a distant sunspot penumbra as seen in AIA 17.1 nm, AIA 160 nm and AIA 9.4 nm channels, outside the FOV of SUSI. The spectral moment anomalies, coronal brightenings and the magnetic configuration support the interpretation of this event as a small-scale microflare-like reconnection burst.

Photospheric magnetic fields triggering an M-class flare observed by SUNRISE III

*Ryohtaroh T. Ishikawa[1], Yukio Katsukawa[2], Masahito Kubo[2], Yusuke Kawabata[2], Kanya Kusano[3], Yumi Bamba[4], Takayoshi Oba[5], Takuma Matsumoto[3], Yoshihiro Naito[6,2], Toshifumi Shimizu[7], Hirohisa Hara[2], Fumihiro Uraguchi[2], Toshihiro Tsuzuki[2], Kazuya Shinoda[2], Tomonori Tamura[2], Yoshinori Suematsu[2], Carlos Quintero Noda[8], Jose Carlos del Toro Iniesta[9], David Orozco Súarez [9], Andreas Korpi-Lagg[10]

1. National Institute for Fusion Science, 2. NAOJ, 3. Nagoya Univ., 4. National Institute of Information and Communications Technology, 5. Kanazawa Univ., 6. SOKENDAI, 7. ISAS/JAXA, 8. IAC, 9. IAA-CSIC, 10. MPS

We investigated an M5.3-class flare that occurred on 13 July 2024, observed by SUNRISE III. The FOV of Sunrise Chromospheric Infrared spectro-Polarimeter (SCIP) covered a part of the polarity inversion line in the active region, and complex Stokes V profiles appeared in the photospheric Fe I lines. The Stokes inversion found that these profiles can be explained by a two-component atmosphere with both magnetic polarities present within the spatial resolution element. In contrast, the antisymmetric Stokes V profile of the chromospheric Ca II line can be explained by a single magnetic component. These differences in the Stokes V profiles suggest a bald patch structure in the photosphere on the polarity inversion line. It is inferred that magnetic reconnection occurred at this bald patch and triggered the flare.

HINODE-SOT at 20 Years: Long-Term Instrument Status and Lessons from Sustained Solar Observations

*Yukio Katsukawa[1], Joten Okamoto[1], Masahito Kubo[1], Toshifumi Shimizu[2], Marc DeRosa[3], Richard Shine[3], Thomas Cruz[3], HINODE SOT team

1. National Astronomical Observatory of Japan, 2. ISAS/JAXA, 3. Lockheed Martin

Since its launch in 2006, the Solar Optical Telescope (SOT) onboard HINODE has continuously delivered high-resolution, seeing-free observations of the solar surface, providing an exceptional opportunity to evaluate how a space-borne solar telescope evolves when operated and maintained over multi-decade timescales. While Filtergraph (FG) observations were terminated in 2016, the Spectro-Polarimeter (SP) has remained in operation and continues to provide precise, stable measurements of photospheric vector magnetic fields. In this contribution, we present an updated status overview of SOT and compile long-term trends in key performance indicators across 20 years of operations, with particular emphasis on telescope thermal conditions, image-stabilization performance, and gradual changes in optical focus and system throughput. Based on these results, we summarize instrument knowledge gained through sustained solar observations and outline practical operational lessons learned, which are expected to be directly relevant to the design and long-term planning of future solar observing instruments that require stable performance and reliable calibration over decades.

Two-Fluid Atmospheric Response to Collisional Return-Current Transport in Solar Flares

*Meriem Alaoui[1], Jeffrey Reep[2]

1. IREAP University of Maryland & NASA/GSFC, 2. University of Hawaii

Flare hydrodynamic models often use electron-beam parameters constrained by hard X-ray observations, but the inferred energy deposition depends on how return-current losses are treated.

We present single- and two-fluid HYDRAD simulations driven by self-consistent beam and return-current transport using RunRC. RunRC provides a 1D treatment of beam propagation, Coulomb losses, return-current losses, and quasi-collisionless transport, allowing us to select hydrodynamic cases in which the return-current heating term is physically justified.

In the collisional regime, which is the focus of this talk, return-current losses redistribute the beam energy deposition relative to Coulomb collisions alone. The magnitude and location of this redistribution are sensitive to beam flux density and the initial atmospheric stratification: return currents heat the corona faster and to higher temperatures, while reducing the beam energy that reaches the chromosphere during the early impulsive phase.

Coulomb-collision-only runs show little sensitivity to the single- versus two-fluid treatment, while return-current-plus-Coulomb runs diverge because the return-current electric field and Joule heating depend on the evolving electron temperature. In two-fluid runs, electron-ion thermal decoupling modifies conductive fluxes and both the coronal and lower-atmosphere/TR response.

These results provide a more self-consistent basis for interpreting Hinode/EIS and other EUV spectroscopic diagnostics of where flare energy is deposited and how the lower atmosphere responds.

Highlights from Hinode/SOT as it Turns 20!

*Marc DeRosa[1], Yukio Katsukawa[2], Joten Okamoto[2], Sanjiv Tiwari[3,1], Rebecca Centeno[4]

1. Lockheed Martin Solar and Astrophysics Laboratory, 2. National Astronomical Observatory of Japan, 3. SETI Institute, 4. High Altitude Observatory (NSF NCAR)

At this meeting, we celebrate the Hinode spacecraft's 20 years in operation. Hinode relies on its three coordinated instruments to trace the flow of energy from the solar photosphere into the corona. The Solar Optical Telescope (SOT) has captured both narrow- and broad-band filtergrams of the solar photosphere and chromosphere (from the SOT-Filtergraph, up to 2016), plus polarizaton spectra that yield vector magnetic field measurements of the solar photosphere (from the SOT-Spectro-polarimeter). Over the past two decades, the data provided by SOT has been instrumental in groundbreaking solar research, including mapping the evolution of quiet-Sun magnetic fields, tracking the dynamics of near- and off-limb spicules, analyzing long-term changes in the Sun's polar magnetic fields, observing material flows in filaments and prominences, investigating magnetic flux emergence from the solar interior to the photosphere, and studying fine-scale structure dynamics within magnetic plage and sunspots. These observations complement observations of the overlying corona by XRT and EIS on Hinode, enabling the dynamics to be tracked through the solar photosphere and chromosphere and into the corona. In this contribution, we highlight research topics for which SOT played a key role in investigating and understanding.

The wide field of view magnetic field measurements of multi-temperature plasma with a large-aperture infrared birefringent tunable filter and DKIST

*Shin'ichi Nagata[1], Ayumi Asai[1], Denis P Cabzezas[1], Tomoya Iju[1], Goichi Kimura[1], Satoru Ueno[1], Yoshikazu Nakatani[1], Takaaki Yokoyama[1]

1. Kyoto University

To understand the fast magnetic reconnection in the solar flares and MHD instabilities of the magnetic flux ropes and the relationship between them, accurate magnetic field measurements with high special and temporal resolution (0.5 arc-sec and 10 sec) and a large field of view (2 arc-min square) above the photosphere are required; imaging spectro-polarimetric observations of He I 1080 nm (chromosphere and filaments, prominences) and Fe XIII 1074 nm (a million degree corona above the limb) at state of the art ground-based solar telescope like Daniel K. Inouye Solar Telescope (DKIST) can provide such key measurements. In this poster, we show a system that uses a four-stage Lyot filter and a two-stage Michelson filter. The 50 mm clear aperture is sufficient to observe the 120 arc-sec field of view at DKIST. Using these calcite blocks, a bandwidth of 0.08 nm for the FeXIII 1075 nm line can be achieved with a Lyot filter, and a bandwidth of 0.02 nm for the HeI 1083 nm line can be achieved by adding a Michelson to the Lyot filter. A polarizing beam splitter placed at the exit of the Lyot filter enables simultaneous dual-wavelength observation.

Hinode Observations of Solar Polar Magnetic Field Evolution across Polarity Reversals in Cycles 24 and 25

*Daikou Shiota[1,2], Masahito Kubo[3], Yukio Katsukawa[3], Arisa Fujimori[3], Masumi Shimojo[3], Haruhisa Iijima[2], Takuma Matsumoto[2], Satoshi Masuda[2]

1. National Institute of Information and Communications Technology (NICT), 2. Institute for Space-Earth Environmental Research, Nagoya University, 3. National Astronomical Observatory of Japan

Observations of magnetic fields in the solar polar regions play an important role in understanding the long-term variation of solar magnetism. Furthermore, fast solar wind emanates from large coronal holes formed in the polar regions around solar minima. Understanding variations in the solar polar magnetic field is also an important element for improving the accuracy of space weather forecasts.
The first Hinode SOT-SP observations of the polar regions revealed the fine structure of photospheric vector magnetic fields of the polar areas: the presence of many patchy magnetic concentrations with intrinsic field strengths of over 1 kG distributed across the entire polar region (Tsuneta et al. 2008) around the solar minimum. Following the first polar observation, monitoring of the polar regions has continued for 19 years. Since 2012, the monitoring of the entire polar regions has been conducted as HOP 206: Periodic SOT-SP observations during a month at the proper timing (March for the south pole and September for the north pole). The monitoring covers the period longer than one solar cycle and revealed that the distributions of large magnetic patches show variations associated with the solar activity cycle while small magnetic patches do not change as reported in the previous study (Shiota et al 2012).
In this paper, we present a summary of HOP 206 observations of long-term variation of magnetic fields in both solar polar regions. In addition, we find that polarity reversal proceeds non-uniformly in longitude, with new-polarity flux accumulating in specific longitudinal sectors, which are associated with the formation of coronal holes.

High-order Paschen emission from the quiet-Sun off-limb chromosphere.

*Haocheng Yu[1]

1. National Astronomical Observatory of Japan

We report the detection of high-order hydrogen Paschen emission lines (Pa~15, Pa~16, and Pa~17) in the quiet-Sun chromosphere off the solar limb using the Chromospheric Infrared SpectroPolarimeter (SCIP) on board the {\sc Sunrise~iii} balloon telescope. These lines reveal thread-like structures resembling spicules and exhibit systematically smaller Doppler velocities than Ca~II~854.2~nm, indicating that they are optically thinner and more affected by line-of-sight averaging, especially near the limb. Non-LTE radiative transfer synthesis using the spherical code \texttt{rhsphere} reproduces the overall spectral properties and supports a recombination-driven formation mechanism. The observed ratios among the three Paschen lines show significant dispersion and systematic deviations from spherically symmetric one-dimensional model predictions. These results suggest that multidimensional plasma structure and inhomogeneity influence the formation of high-order Paschen lines. The study demonstrates the potential of these lines as a new probe of optically thin, recombination-dominated plasma in the off-limb chromosphere.

Advancing Space Plasma Physics and Space Weather Research through Continuous Solar X-ray Focusing Imaging Spectroscopy with a CubeSat

*Noriyuki Narukage[1]

1. National Astronomical Observatory of Japan

The Sun is an active star filled with plasma and magnetic fields, where energy release occurs across broad temporal and spatial scales. Solar flares, coronal heating, solar wind acceleration, and particle acceleration are key processes for understanding high-energy plasma phenomena. Solar activity can also disturb the near-Earth space environment and affect satellites, communications, navigation systems, and power grids. Understanding solar activity is therefore important for both space plasma physics and space weather research. Dynamic solar phenomena produce hot plasma and non-thermal particles that emit X-rays. X-ray focusing imaging spectroscopy, which measures the position, time, and energy of individual photons, is a powerful tool for diagnosing these processes. We have demonstrated the world’s first focusing X-ray imaging spectroscopy of the Sun through the FOXSI sounding rocket series, including solar flare observations. However, sounding rockets provide only about five minutes of observing time, preventing continuous coverage of key phases such as early flare evolution. This research will develop a CubeSat platform for continuous solar observation with a compact X-ray telescope and a high-speed X-ray camera. The X-ray mirrors provide spatial resolution and dynamic range, while the camera provides energy and temporal resolution. The launch is planned for 2030, followed by one year of continuous observations. Through this mission, we aim to realize the world’s first continuous soft X-ray focusing imaging spectroscopy of the Sun. The observations will provide X-ray spectra resolved in space and time, enabling studies of solar flares, giant arcade structures, active regions, quiet-Sun regions, and fast solar wind source regions. They will address impulsive energy release, plasma heating, plasma acceleration, and particle acceleration, while supporting space weather research. The data and analysis software will be released to promote collaboration, community building, and future large-scale solar missions.